Debt Capacity and Slow Deleveraging with Partial Default*

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Abstract

Sovereign defaults are rarely complete: governments withhold part of their debt service, the missed payments accrue as arrears, and debt keeps growing while the sovereign stays in the market. Arellano, Mateos-Planas, and Ríos-Rull (2023) built a quantitative theory of *partial default* around these facts. We study their model in continuous time, where the absence of a stopping problem makes it tractable, and characterize what the structure of the default costs implies for debt. For any productivity process, wherever the restriction on new issues introduced below is slack, the default decision is static and its thresholds are explicit in the state and the bond price. In the deterministic model, the sovereign's debt capacity, the largest debt it will service in full at the risk-free price, is a closed-form function of the default costs, the recovery on arrears and the maturity of the debt, independent of its impatience and of its risk aversion; the first default is a jump whose size the cost parameters alone determine; and above capacity the sovereign deleverages slowly, at a speed that vanishes like the cube root of the excess debt, while the bond price leaves par with a vertical tangent. The same model admits self-fulfilling partial default: a pessimistic price makes withholding cheaper, the sovereign borrows into a permanent partial default, and the arrears it accumulates validate the price. For a risk-neutral sovereign we characterize when this happens: the impatience wedge must exceed a threshold that, when the default cost is not too convex, is the spread implied by the first default. Calibrated to the targets of the discrete-time model, the diffusion model fits them at least as well, with closed-form thresholds, and shows the sovereign remaining a net issuer for years into a default episode. The computation needs no smoothing shocks and delivers exact long-run distributions and impulse responses. The model also needs an explicit borrowing limit at high debt, where a sovereign in full default would otherwise borrow at any positive price; an underwriting standard on new issues makes the level of that limit irrelevant.

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1 Introduction

Sovereign defaults are rarely complete. Governments in distress typically continue to service some of their obligations while suspending payments on others, missed payments accumulate as arrears that add to the stock of debt, and countries frequently emerge from a default episode owing more than when it began. Arellano, Mateos-Planas, and Ríos-Rull (2023) document these facts and build a theory of *partial default* around them: a sovereign that chooses, period by period, what share of its debt service to withhold, at a cost that rises with the share withheld. Their framework replaces the binary default decision of the standard model, introduced by Eaton and Gersovitz (1981) and quantified by Aguiar and Gopinath (2006) and Arellano (2008), with a continuous one, and it replaces exclusion from financial markets with continued access at prices that reflect the expected intensity of default.[1] In their data emerging economies are in partial default about a third of the time and withhold about forty percent of the debt service when they are, and their calibrated model reproduces the frequency, the depth and the dynamics of these episodes. We regard partial default as the right description of sovereign distress and the natural baseline for the quantitative theory of sovereign debt.

This paper formulates the partial default model in continuous time and characterizes its equilibria. The formulation is natural, since the default intensity is a flow decision and arrears accrue continuously, and it is tractable in a way the discrete-time model is not: because the sovereign never loses market access there is no stopping problem, a single Hamilton–Jacobi–Bellman (HJB) equation for the value and a single linear equation for the bond price describe the equilibrium at every state, and the default decision, given the price, is a static maximization. What this buys is a set of results on how the structure of the default costs shapes the sovereign's debt, results that the discrete-time simulations contain but do not isolate. We organize them as three contrasts with the standard model of binary default: debt capacity, the adjustment above capacity, and the multiplicity of equilibria.

The first concerns debt capacity. In the binary model the debt a sovereign can carry is the outcome of the whole dynamic problem, and it is known only numerically. Under partial default it is a static comparison. Withholding a unit of debt service raises consumption by one minus the market value of the arrears it creates: it is the issuance of debt to existing creditors at an implicit price that exceeds the market price and does not depend on it, paid for with the output cost of default, which makes exact the description by Arellano, Mateos-Planas, and Ríos-Rull (2023) of partial default as a flexible way to raise funds (Lemma 1). Given debt, productivity and the price, the share withheld maximizes a concave resource function and depends neither on the value function nor on the direction in which debt is moving (Proposition 1), wherever the underwriting standard introduced below is slack, so the regions of no, partial and full default

1. Textbook treatments of the standard framework are Uribe and Schmitt-Grohé (2017) and Aguiar and Amador (2021); Aguiar et al. (2016) survey its quantitative extensions. In models with renegotiation, following Yue (2010) and Pitchford and Wright (2012), the sovereign returns to the market with a debt level set by bargaining, as in Mihalache (2020) and Dvorkin et al. (2021).

are explicit in the state and the price (Proposition 2). In the deterministic model, the sovereign borrows at the risk-free price up to the largest debt it is willing to service in full and stops there, so that its debt capacity is the no-default threshold at the risk-free price (Theorem 1): a closed-form function of the default costs, of the recovery on arrears and of the maturity of the debt, which does not depend on the sovereign's impatience, on its risk aversion or on the borrowing limit. The formula says what raises capacity: a steeper variable cost and a larger fixed cost, a higher recovery on arrears, which makes withholding more expensive, and longer maturities. It also says that a sovereign never defaults a little. The first default is a jump whose size the cost parameters alone determine, half of the debt service at the calibrated parameters, because a small default is not worth its fixed cost.

The second concerns what happens above capacity. In the binary model a sovereign past its limit defaults, and the adjustment is a jump into exclusion or renegotiation. Under partial default it is gradual. Above its capacity the sovereign is in partial default, the price is below par, and it repays, but at a speed that vanishes like the cube root of the excess debt, while the price leaves par with a vertical tangent (Theorem 2). A sovereign slightly over its limit stays there for a long time and works its debt down slowly, and even a small excess is visibly priced. The law has a two-term expansion whose constants are explicit in the primitives, and the computed solution follows it (Figure 4). From below, the sovereign approaches its capacity along a square-root law, in finite time (Lemma 2). The value is differentiable at capacity but not twice, so a scheme that differences it there sees a kink that is not one. The equilibrium with these properties exists and is unique in its class whenever the branch that leaves capacity along the cube-root law stays regular, a condition on an explicit differential equation (Proposition 11).

The third is multiplicity. Partial default admits a self-fulfilling equilibrium of a particular form. If lenders expect the sovereign to settle at a higher debt and to withhold a constant share of its debt service there forever, they price the bond accordingly, and at the lower price withholding is cheaper, so the sovereign starts defaulting at debt levels at which it would otherwise have repaid, and borrows past its old capacity, because the default is priced and the funds it raises cost it only the risk-free rate. It drifts into a permanent partial default over years, through arrears, and the arrears validate the price. Nothing happens suddenly: there is no run and no loss of market access. The candidates form a continuum indexed by the intensity of the permanent default, and whether one is an equilibrium is a two-point problem, the sovereign's willingness to borrow into default at a discount all the way from zero debt. For a risk-neutral sovereign we nearly solve it: a self-fulfilling partial default can be reached from zero debt only if the gain from front-loading the borrowed funds covers the output cost of the default, and it is an equilibrium when, in addition, the steady state attracts and the price schedule continues regularly above it; when the default cost is not too convex, the threshold is that the impatience wedge exceed the spread implied by the first default (Proposition 13). With curvature the threshold is higher in every case we computed, and it falls to the risk-neutral one as the curvature vanishes. This is the slow-moving crisis of Lorenzoni and Werning (2019) with arrears in the place of rollover, and

the dilution of Aguiar and Amador (2020) with a continuous default decision; an underwriting standard on new issues rules it out whenever the spread implied by the first default exceeds the spread at the floor (Proposition 12), which it does, narrowly, at the calibrated parameters.

The quantitative section calibrates the diffusion model to the eleven moments that Arellano, Mateos-Planas, and Ríos-Rull (2023) target, with their free parameters and their weights, and finds that it fits them at least as well under their own loss. The thresholds and the debt capacity of the theory are visible in the policies, and the ergodic distribution concentrates on the locus where debt is stationary, which lies below capacity in good times and in the partial-default region in bad times. The response to a negative productivity shock is the episode the data describe: the default intensity jumps and stays elevated for a decade, the spread rises, and debt rises for six years before it is worked off. The decomposition of that rise is exact, and it says that the sovereign in partial default remains a net issuer: new issues add more to the debt than arrears do for five years into the episode. The mechanism is the fixed cost. Outside default the fixed cost is a brake the sovereign will not cross, so it stops at its capacity; once a bad shock has moved the threshold below the current debt, the fixed cost is being paid regardless, the episode will end when productivity recovers rather than when the debt is repaid, and the sovereign's debt target, now set by the dilution of its own price rather than by the fixed cost, is higher. Default is lumpy at entry and sticky thereafter.

Two further contributions are technical. The first is computational. The equilibrium is computed by an implicit upwind finite difference scheme in the tradition of Achdou et al. (2022), with the value and the price updated jointly because the alternating iteration between them is unstable, and the default decision needs no smoothing: the intensity is a continuous control, so the price schedule is continuous wherever the state moves through a point. Discrete-time models with long-term debt rely on i.i.d. shocks, to income or to the default and borrowing choices, to make the price schedule converge (Chatterjee and Eyigungor 2012; Dvorkin et al. 2021; Mihalache 2020); these devices are widely used and effective, yet they carry both a conceptual cost, the smoothed decision is not the one the model describes, and a practical one, convergence that depends on their scale (see Mihalache (2025) for a survey). The stationary distribution and the impulse responses are computed exactly from the generator of the discretized process rather than by simulation, and the same scheme solves the deterministic, chain and diffusion models. The second concerns the model at high debt, where it is not well posed without an explicit borrowing limit. In states where productivity is low enough that default carries no fixed cost, a sovereign in full default pays nothing, bears only the variable cost, and faces a marginal cost of debt close to zero, while lenders still pay a positive price because productivity will recover; it then issues at any positive price up to the borrowing limit that, as in every incomplete-markets model, rules out Ponzi schemes, whatever the level of the limit (Section 3.2). Without the limit debt would not grow forever either, since debt that is never serviced again is worthless (Proposition 7), but whether an equilibrium exists then is open. Quantitative models leave the limit implicit, as the top of the debt grid, so a choice about the computation becomes part of the model. We adopt an

underwriting standard, a lower bound on the price of new issues as in Chatterjee and Eyigungor (2015), under which the full-default region is transient and the level of the borrowing limit is irrelevant (Propositions 9 and 10); where it binds, the sovereign applies its default rule at its own shadow price of debt and defaults more (Proposition 8). At the calibrated parameters the standard binds on less than one percent of the ergodic mass and the accumulation it removes is negligible; at other parameters, documented in Appendix C, the accumulation decides the tails of the computed distribution and the standard is what makes the solution independent of the grid.

Our paper belongs to the recent literature on quantitative sovereign default in continuous time. Bornstein (2020) compares discrete and continuous time versions of the standard model, finds that computation is faster in continuous time, and identifies a *painful deleveraging* effect specific to settings in which the stock of debt is slow-moving.[2] Tourre (2017) studies the macro-financial implications of the standard model with recursive utility and a rich pricing kernel. Hurtado, Nuño, and Thomas (2023) is the paper closest to ours: they study a sovereign with long-term debt denominated in local currency, who can reduce the real value of the debt through inflation. Inflation lowers the real value of the entire stock of debt, whereas partial default applies to the debt service flow and raises the stock through arrears. All three papers solve optimal stopping problems with HJB variational inequalities, because default is a discrete transition into exclusion; the partial default model has no such transition, which is the source of its tractability.

Continuous-time methods have also been at the forefront of theoretical work on sovereign debt. Aguiar and Amador (2020) show that self-fulfilling debt dilution produces multiple equilibria in a continuous-time model with long-term bonds and a binary default decision; the self-fulfilling partial default of Section 3.3 is the same kind of multiplicity with a continuous default decision, where the dilution runs through arrears. Amador and Phelan (2021, 2023) develop the reputational theory of sovereign default, the latter in a setting with partial default, the closest theoretical treatment of a partial default decision we know of. The slow-moving debt crises of Lorenzoni and Werning (2019) are the closest relatives of the self-fulfilling partial-default steady states of Section 3.3: in our model a pessimistic price sustains a higher default intensity which sustains the price, and whether such a steady state can be reached from low debt depends on the sovereign's willingness to borrow into default at a discount. The underwriting standard we adopt is a lower bound on the price of new issues, which Chatterjee and Eyigungor (2015) also impose in their analysis of a seniority arrangement for sovereign debt; Hatchondo, Martinez, and Roch (2022) study a related fiscal rule, a spread brake that caps the deficit when the sovereign spread exceeds a threshold.

On the computational side we follow Achdou et al. (2022) and Fernández-Villaverde, Hurtado, and Nuño (2023). The partial default model requires two additions to their scheme: a joint implicit update of the value and the price, because the alternating iteration between them is

2. Rendahl (2022) discusses the extent to which similar results can be obtained in discrete time by adapting the sparse matrix methods at the core of the continuous-time solution.

unstable, and a monotone treatment of the non-concave Hamiltonian created by the fixed cost of default.

The rest of the paper proceeds as follows. Section 2 presents the model, derives the HJB and pricing equations, defines the equilibrium and discusses the key assumptions, including the borrowing limit and the underwriting standard. Section 3 characterizes the equilibrium: the results for any productivity process first, then the behavior at high debt and what the standard does there, then the deterministic model, where debt capacity, the cube-root law and the self-fulfilling equilibria live, and the chain model, with a numerical illustration of each. Section 4 contains the quantitative analysis of the diffusion model: calibration, computation, policies, the ergodic distribution and the impulse responses. Section 5 concludes. The appendices contain the proofs, the numerical implementation and robustness checks.

2 The Partial Default Model

The economy consists of a small open economy run by a risk-averse sovereign and of a unit measure of competitive, risk-neutral international lenders. Time is continuous. We describe the two problems in turn and define the equilibrium.

2.1 The Sovereign's Problem

The sovereign receives an endowment stream, borrows abroad with long-term bonds, and chooses at every instant what fraction of its scheduled debt service to pay.[3] The part of the debt service it does not pay accrues as arrears and adds to the stock of debt.

Preferences and endowment. The sovereign maximizes

$$\mathbb{E}_0 \int_0^\infty e^{-\rho t}\, u(c_t)\, dt,$$

where ρ is its discount rate and u is increasing and strictly concave. Log productivity z_t follows a Markov process with generator $\mathcal{L}_z$, which we leave general in this section: in Section 3 it is constant or a finite-state chain, in Section 4 an Ornstein–Uhlenbeck diffusion. We call the model with a general Markov process the *general model*, and its three special cases the *deterministic model* (constant productivity), the *chain model* (a finite-state Markov chain) and the *diffusion model* (an Ornstein–Uhlenbeck diffusion). Output is e^{z_t} when the sovereign is current on its debt; when it defaults, output is $\phi(d_t, z_t)\, e^{z_t}$, where the cost function ϕ is described below.

3. We maintain the standard assumptions of centralized borrowing and centralized default, so that we do not make explicit the private sector and the policy instruments through which the sovereign implements its preferred outcomes. For the case of decentralized borrowing see Kim and Zhang (2012).

Debt. The stock of debt is B_t. Each unit of debt pays a coupon λ and matures at rate δ, so that the scheduled debt service is $(\delta+\lambda)B_t$ per unit of time.[4] The sovereign chooses a *default intensity* $d_t \in [0,1]$, the share of the scheduled debt service it withholds. Of every unit withheld, a fraction κ is added to the stock of debt as arrears; $1-\kappa$ is the haircut suffered by lenders on defaulted payments. The sovereign also issues ℓ_t units of new debt at the market price q_t (when $\ell_t < 0$ it buys debt back). The budget constraint and the law of motion of debt are

$$c_t = \phi(d_t, z_t)\, e^{z_t} - (1-d_t)(\delta+\lambda)B_t + q_t\, \ell_t, \tag{1}$$

$$\dot{B}_t = \ell_t - \delta B_t + \kappa(\delta+\lambda)\, d_t\, B_t. \tag{2}$$

The three terms in (1) are income net of the default cost, the part of the debt service that is paid, and the proceeds from issuance; the three terms in (2) are issuance, amortization and the accrual of arrears. It is convenient to write

$$a(d) := \delta - \kappa(\delta+\lambda)\, d, \qquad a := a(1) = \delta - \kappa(\delta+\lambda), \tag{3}$$

so that $\dot{B}_t = \ell_t - a(d_t)B_t$: absent issuance, debt decays at rate $a(d)$, which is lowest in full default. We assume $a > 0$, so that even a sovereign in full default sees its debt shrink, slowly, when it does not issue. Debt is bounded, $B_t \in [0, \bar{B}]$: the sovereign does not lend, and, as in every incomplete-markets model, a borrowing limit $\bar{B}$ rules out Ponzi schemes. Quantitative default models usually leave the limit implicit, as the top of the grid on which debt is computed; we make it explicit because, without the underwriting standard of Section 2.4, its level matters.

The cost of default. Following Arellano, Mateos-Planas, and Ríos-Rull (2023) the cost of default has a variable and a fixed component,

$$\phi(d,z) = \left(1-\gamma_0 d^{\gamma_1}\right) F(z) \;\text{ for } d>0, \qquad \phi(0,z)=1, \qquad F(z) = 1-\gamma_2\left(e^z - y^*\right)\mathbb{1}\{e^z \geq y^*\}, \tag{4}$$

with $\gamma_0 \in (0,1)$, $\gamma_1 > 1$ and $\gamma_2 \geq 0$. The variable cost $\gamma_0 d^{\gamma_1}$ is convex in the default intensity with zero marginal cost at $d=0$. The fixed cost $(1-F(z))e^z$ is paid whenever the sovereign defaults at all, however little, and only when output e^z exceeds the threshold y^*; it is larger in better times. Below the threshold default carries only the variable cost, and we call these states the *free-default region*. The jump of ϕ at $d=0$ above the threshold is what creates a region of no default: a small default is not worth its fixed cost.

4. This is the continuous-time counterpart of the long-term bond of Hatchondo and Martinez (2009) and Chatterjee and Eyigungor (2012): the maturing fraction δ of the stock is repaid at par, and a single state variable summarizes the debt.

The drift of debt. Substituting ℓ_t out of (1) using (2), the drift of debt is a function of the state, the price and the two controls,

$$S(B,z,c,d,q) = \frac{c - \phi(d,z)e^z + (1-d)(\delta+\lambda)B}{q} - a(d)\,B, \tag{5}$$

and we treat consumption c and the default intensity d as the controls. Equivalently, for a given drift S,

$$c = \mathcal{R}(d;B,z,q) + q\,S, \qquad \mathcal{R}(d;B,z,q) := \phi(d,z)e^z - (1-d)(\delta+\lambda)B + q\,a(d)\,B. \tag{6}$$

We call $\mathcal{R}$ the *resource function*: it is consumption at zero drift, and it is all that the default decision affects for a given drift. The characterization of Section 3 rests on this decomposition.

Arrears as borrowing. The decomposition has a second reading, which makes exact the description of partial default by Arellano, Mateos-Planas, and Ríos-Rull (2023) as a flexible way to raise funds. Let $n_t := \ell_t + \kappa(\delta+\lambda)d_tB_t$ be the total addition to the stock of debt, from issuance and from arrears, so that $\dot{B}_t = n_t - \delta B_t$. Substituting into (1),

$$c_t = \phi(d_t,z_t)e^{z_t} - (\delta+\lambda)B_t + q_t\,n_t + (1-\kappa q_t)(\delta+\lambda)B_t\,d_t. \tag{7}$$

Holding the path of debt fixed, withholding a unit of debt service raises consumption by $1-\kappa q_t > 0$: the unit saved, net of the market value of the κ units of arrears that replace market issuance one for one. Equivalently, arrears are the issuance of κ units of debt to existing creditors at the implicit price $1/\kappa$, which exceeds the market price and does not depend on it, paid for with the output cost $(1-\phi)e^z$ of default. Section 3.1 draws the consequences.

The HJB equation. Given a bond price function $q(B,z)$, the sovereign's value $V(B,z)$ solves

$$\rho V(B,z) = \max_{c,\ d\in[0,1]} \Big\{ u(c) + S(B,z,c,d,q(B,z))\,V_B(B,z) + \mathcal{L}_z V(B,z) \Big\}, \tag{8}$$

subject to the state constraints $0 \le B \le \bar{B}$ and the underwriting standard of Section 2.4. Subscripts denote partial derivatives. The derivation is in Appendix A. Unlike models with a binary default decision, there is no variational inequality and no second regime: the sovereign never loses market access, and a single value function with possibly discontinuous policies describes it at all states.

First-order conditions. At an interior drift, consumption satisfies

$$u'(c)\,q(B,z) = -V_B(B,z), \tag{9}$$

the price of a bond times the marginal utility of the resources it brings equals the marginal cost of debt in utility units. By (6) the default intensity enters the objective only through $\mathcal{R}$, which is strictly concave on $(0,1]$ with

$$\mathcal{R}_d(d) = -\gamma_0\gamma_1 F(z)e^z d^{\gamma_1-1} + (1-\kappa q)(\delta+\lambda)B.$$

The marginal gain from withholding a unit of debt service is $(1-\kappa q)(\delta+\lambda)B$: the payment saved, net of the market value κq of the arrears it creates. The interior maximizer is

$$d_{\text{int}}(B,z,q) = \min\Big\{1,\ \Big[\frac{(1-\kappa q)(\delta+\lambda)B}{\gamma_0\gamma_1 F(z)e^z}\Big]^{1/(\gamma_1-1)}\Big\}, \tag{10}$$

and the optimal default intensity is either 0 or d_{int}, whichever gives the larger resource function. Section 3.1 makes this precise and shows that it holds for every maximizer of (8).

2.2 International Lenders

A unit measure of deep-pocketed, competitive, risk-neutral lenders with outside option r buy and hold the sovereign's bonds. A unit of debt held at t yields the flow $(1-d_t)(\delta+\lambda)$ in cash and $\kappa(\delta+\lambda)d_t$ in new units (the arrears), and δ of it matures. The break-even price is the expected discounted value of the cash flow, where the discount rate is adjusted for the growth of the position through arrears,

$$q(B,z) = \mathbb{E}_{B,z}\int_0^\infty \exp\Big(-\int_0^t \xi(d_s)\,ds\Big)\,(1-d_t)(\lambda+\delta)\,dt, \qquad \xi(d) := r+\delta-\kappa(\delta+\lambda)\,d. \tag{11}$$

The effective discount rate $\xi(d)$ combines the lenders' opportunity cost r, the amortization δ, and the accrual of arrears, which lowers it. By the Feynman–Kac formula, q solves the linear equation

$$\xi\big(d(B,z)\big)\,q(B,z) = (1-d(B,z))(\lambda+\delta) + \tilde{S}(B,z)\,q_B(B,z) + \mathcal{L}_z q(B,z), \tag{12}$$

where $\tilde{S}(B,z)$ is the equilibrium drift of debt and $d(B,z)$ the equilibrium default policy. The evolution of the stock of debt is governed by $\tilde{S}$, which includes issuance; the cash flow of a single unit is governed by $\xi(d)$, which does not.

Spreads and duration. The price of a bond that never defaults, the *risk-free price*, is $q^{\text{rf}} = (\delta+\lambda)/(r+\delta)$. We set the coupon equal to the risk-free rate, $\lambda = r$, so that $q^{\text{rf}} = 1$ and the face value of a bond equals its risk-free value; we keep the symbol q^{rf} wherever a price is the risk-free one. The yield to maturity implied by a price q is $(\delta+\lambda)/q - \delta$, so the spread over the risk-free yield is

$$\text{spread}(B,z) = \frac{\delta+\lambda}{q(B,z)} - (\delta+r), \tag{13}$$

and the risk-free Macaulay duration of the bond is $1/(\delta+r)$.

2.3 Equilibrium and the Ergodic Set

Definition 1. A *Markov equilibrium* consists of a value function $V(B,z)$, policies $c(B,z)$ and $d(B,z)$, and a bond price $q(B,z)$ such that

1. given q, the value V solves the HJB equation (8) and (c,d) attain the maximum on its right-hand side, subject to the state constraints $0 \leq B \leq \bar{B}$ and the underwriting standard of Section 2.4;

2. given the policies, q solves the pricing equation (12), equivalently (11).

The stationary distribution. Let $g(B,z)$ be the density of the state. Given the equilibrium drift $\tilde{S}$ and the generator $\mathcal{L}_z$ with adjoint $\mathcal{L}_z^*$, the density solves the Kolmogorov forward equation

$$\partial_t g = -\partial_B \left[\tilde{S}(B,z)\, g\right] + \mathcal{L}_z^* g, \tag{14}$$

and the stationary distribution solves $0 = -\partial_B[\tilde{S}g] + \mathcal{L}_z^* g$. The forward equation is the adjoint of the HJB equation: in the discretized model the same sparse matrix that solves the HJB equation, transposed, gives the stationary distribution, so that the long-run moments are available without simulation (Appendix B). The stationary distribution can have atoms: at $B = 0$ when the sovereign repays in full at high productivity, and at stationary points of debt in the deterministic and chain models. We call its support the *ergodic set*; its shape, in particular how far it extends into the default region, is a central object of the paper.

2.4 Discussion

Comparison with the discrete-time model. The model follows Arellano, Mateos-Planas, and Ríos-Rull (2023) in letting the sovereign withhold a share of the scheduled payments at a cost that is convex in the share, with a fixed cost that depends on productivity. Three points of comparison are worth noting. First, the discrete-time model distinguishes between the debt due in the period and the stock of debt; in continuous time the debt due is the flow $(\delta+\lambda)B$ and the two coincide up to the factor $\delta+\lambda$. Second, the sovereign in our model can buy back debt ($\ell < 0$); this matters for the monotonicity of the value in debt, which we prove with buybacks allowed. Third, as in their model but unlike the standard model with a binary default decision, the sovereign never loses market access as a punishment for default: the price at which it can issue falls with the expected default intensity, which is the only punishment through markets. There is therefore a single value function, no stopping problem, and no variational inequality, which is what makes the default decision static (Proposition 1) and the characterization of Section 3 possible. The underwriting standard introduced below does not change this: new placements stop when the price falls below a floor, but the debt continues to trade and the sovereign can buy it back.

The recovery parameter. The law of motion (2) treats a missed payment as a claim that existing creditors hold at face value, of which the fraction κ is expected to survive: $\kappa < 1$ summarizes the haircut that past-due amounts carry in an eventual restructuring, and the arrears rank with the rest of the stock, so they are priced by the market through $q(B, z)$ like every other unit of debt. What the contract fixes is the exchange ratio, one unit withheld for κ units of claims; what the market sets is the value of those claims. The reading of Section 2 is therefore an accounting identity, and its economic content is that the implicit price of this issuance, $1/\kappa$, exceeds the market price and does not respond to it: a sovereign in distress has a source of funds whose terms do not deteriorate with its creditworthiness, which is what makes arrears attractive precisely when the market price is low. We take κ as a parameter, as Arellano, Mateos-Planas, and Ríos-Rull (2023) do. Making it the outcome of a renegotiation would make the implicit price depend on the state and is a natural extension; the results of Section 3.1 hold for any κ in $(0, 1)$ satisfying Assumption 1(i) and would carry over to a κ that depends on productivity.

The underwriting standard. Without a further restriction the borrowing limit is not innocuous: in the states where default carries no fixed cost, a sovereign in full default faces almost no marginal cost of debt while lenders still pay a positive price, because productivity will recover, so, as we argue in Section 3.2, the sovereign issues at any positive price and borrows up to the limit, whatever its level. We therefore impose an *underwriting standard*: the banks that underwrite the sovereign's issues in the financial centers do not bring a new issue to market at a price below a floor $\underline{q}$,

$$\ell_t \leq 0 \quad \text{whenever} \quad q(B_t, z_t) < \underline{q}. \tag{15}$$

The restriction is on placements, not on holdings: the outstanding bonds continue to trade at the price q that competitive lenders set, buybacks remain allowed, and partial default and the accrual of arrears are unchanged. A lower bound of this kind on the price of new issues appears in Chatterjee and Eyigungor (2015). In yield terms, with $\lambda = r$ the floor corresponds to a spread of $(\delta + \lambda)(1/\underline{q} - 1)$; for the value $\underline{q} = 0.6$ we use, this is about 1,280 basis points at the calibrated maturity rate. The level is a judgment: at the calibrated parameters it lies far above the highest price at which the unconstrained sovereign issues in full default, about 0.3, and it binds on less than one percent of the ergodic mass; at the first calibration, where the accumulation is stronger, a floor of 0.6 removes most but not all of it (Appendix C). The restriction is on the price rather than on the quantity of debt because the accumulation is located by the price: it begins at the full-default line in the states where default carries no fixed cost, at debt levels that differ across states, and a floor on the price is a state-dependent debt limit expressed in the one variable both parties observe, the underwriters' counterpart of the spread brakes that Hatchondo, Martinez, and Roch (2022) study as fiscal rules. Section 3.2 shows what the standard does: it makes the full-default region transient and the level of the borrowing limit irrelevant once the limit exceeds the full-default thresholds, it bounds the support of debt, and where it binds it makes the sovereign apply its default rule at a shadow price below the market price. It binds only in deep distress: at

the calibrated parameters, on just under one percent of the ergodic mass, all of it more than one standard deviation below mean productivity.

No trade at a zero price. A second, milder convention rules out the autarky equilibrium in which lenders expect full default forever, the price is zero, and the sovereign keeps itself above the full-default line by issuing at a zero price. We impose that no bonds change hands at a zero price; Section 3.1 shows that the price is then strictly positive at every state.

3 Characterization

This section collects what can be established analytically. Section 3.1 contains the results for the general model, any Markov productivity process: the arrears identity, the static default decision and the thresholds, the bond price and the value. Section 3.2 turns to high debt, where the model needs the borrowing limit and the underwriting standard. Section 3.3 specializes to the deterministic model, where the paper's main results live: debt capacity in closed form, the laws of motion on either side of it, and self-fulfilling partial default. Section 3.4 treats the chain model. Every formal result states the model it covers and whether it assumes the underwriting standard; proofs are in Appendix A. Everything else in this section is one of two other kinds of statement, marked as such: an argument, which explains a mechanism without proving it, or a numerical finding, which names the model and the parameters it was computed at. The numerical illustrations of the deterministic and chain models borrow the parameters of the diffusion model's calibration (Table 1), with productivity held at its mean or at a few fixed levels; they show the mechanics and are not calibrated in their own right, and we refer to them as *the illustration*. Two other parameterizations appear: the *first calibration* of the diffusion model, without the standard (Appendix C), and an *alternative parameterization* with a small first default (Figure 1). Throughout we maintain the following.

Assumption 1.

(i) $\xi(1) = r + \delta - \kappa(\delta + \lambda) > 0$ and $a = \delta - \kappa(\delta + \lambda) > 0$.

(ii) $\lambda = r$, so that $q^{\text{rf}} = 1$; $\rho > r$.

(iii) u is C^2, strictly increasing and strictly concave; $\gamma_1 > 1$, $0 < \gamma_0 < 1$ and $F(z) > 0$.

(iv) The drift is bounded, $|S| \leq \bar{S}$, consumption is bounded below, $c \geq c_{\min} > 0$, and $B \in [0, \bar{B}]$.

Open questions. Most of these results are simple once the model is written in continuous time, which is the point of writing it that way. Three questions remain open, and they are where the theory of this model is hard: existence of the coupled equilibrium in the chain and diffusion models, since the best response from prices to prices is neither continuous nor monotone; its uniqueness; and whether self-fulfilling partial-default steady states exist. For the last we have

answers in the deterministic model only: a necessary condition under the standard (Proposition 12), and, for a risk-neutral sovereign, conditions that are necessary and, up to an attraction and a continuation condition, sufficient (Proposition 13); with curvature we have numerical findings only.

3.1 Results for any Markov productivity process

Lemma 1 (arrears are borrowing at a fixed price)**.** *In the general model, for any path of total new debt n_t, consumption satisfies (7): a unit of debt service withheld is κ units of debt issued to existing creditors at the price $1/\kappa$, independent of the market price, in exchange for the output cost of default. In particular, for a given drift of debt the default intensity affects consumption only through the resource function $\mathcal{R}(d;B,z,q) = \phi(d,z)e^z - (1-d)(\delta+\lambda)B + q\,a(d)B$, and $\mathcal{R}_d = -\gamma_0\gamma_1 F(z)e^z d^{\gamma_1-1} + (1-\kappa q)(\delta+\lambda)B$ for $d > 0$.*

The lemma is an identity, and it is the identity that organizes the model: the sovereign has two ways of raising funds, the market at the price q and arrears at the price $1/\kappa$, and the second is always the cheaper per unit of debt and always costs output. Everything below follows from it, starting with the fact that the choice between the two is static; the proof of that fact is immediate once the budget constraint is written as in the lemma, which is why we write it that way.

Proposition 1 (the default decision is static)**.** *In the general model, for every state (B,z), price $q \geq 0$ and value function V, every maximizer (d^*, c^*) of the right-hand side of (8) satisfies*

$$d^* \in \arg\max_{d\in[0,1]} \mathcal{R}(d;B,z,q).$$

The default policy depends on $(B,z,q(B,z))$ only, and not on V, on the sign of the drift, or on the constraints on the drift, as long as those constraints do not depend on d.

The qualification matters for the underwriting standard, whose constraint $\ell = S + a(d)B \leq 0$ does depend on d; Proposition 8 in Section 3.2 gives the rule in that case. Everywhere else the default decision is a static maximization of the concave function $\mathcal{R}$, which yields explicit thresholds.

Proposition 2 (default thresholds)**.** *In the general model, fix (B,z,q) with $B > 0$ and $\kappa q < 1$. The optimal default intensity is $d^* = 0$ if the fixed cost exceeds the gain from defaulting,*

$$(1 - F(z))\,e^z \;\geq\; \Gamma(B,z,q) := (1-\kappa q)(\delta+\lambda)B\,d_{\text{int}} - \gamma_0 F(z)e^z d_{\text{int}}^{\gamma_1}, \tag{16}$$

and $d^* = d_{\text{int}}(B,z,q)$ of (10) otherwise. Γ is continuous, strictly increasing in B, strictly decreasing in q, and zero at $B = 0$. Consequently:

(a) there is a threshold $B_{\text{pd}}(z,q)$, nondecreasing in q, with $d^* = 0$ for $B \leq B_{\text{pd}}$ and $d^* = d_{\text{int}} > 0$ for $B > B_{\text{pd}}$;

(b) *in the free-default region* ($F(z) = 1$), $B_{\text{pd}}(z, q) = 0$*: any positive debt is partially defaulted on;*

(c) *if* $F(z) < 1$ *and* $d_{\text{first}}(z) < 1$ *(below), the default intensity jumps at* B_{pd} *from* 0 *to the* first default intensity

$$d_{\text{first}}(z) = \Big[\frac{1 - F(z)}{(\gamma_1 - 1)\gamma_0 F(z)}\Big]^{1/\gamma_1}, \tag{17}$$

which depends on the cost parameters only;

(d) *full default,* $d^* = 1$*, occurs for* $B \geq \max\{B_{\text{fd}}(z, q), B_{\text{pd}}(z, q)\}$ *with* $B_{\text{fd}}(z, q) := \gamma_0\gamma_1 F(z)e^z / \big((1 - \kappa q)(\delta + \lambda)\big)$*; when* $d_{\text{first}}(z) \geq 1$ *the intensity jumps from* 0 *directly to* 1 *at* B_{pd}*;*

(e) *when* $d_{\text{first}} < 1$*, the threshold has the closed form*

$$B_{\text{pd}}(z, q)^{\frac{\gamma_1}{\gamma_1 - 1}} = \frac{\gamma_1}{\gamma_1 - 1} \frac{(1 - F(z))\, e^z\, G^{\frac{1}{\gamma_1 - 1}}}{K^{\frac{\gamma_1}{\gamma_1 - 1}}}, \qquad G := \gamma_0\gamma_1 F(z)e^z, \quad K := (1 - \kappa q)(\delta + \lambda). \tag{18}$$

Two monotonicity properties follow by reading the formulas: $d^*(B, z, q)$ is nondecreasing in B and nonincreasing in q, in γ_0, in κ, and in $F(z)e^z$ holding the fixed cost $(1 - F(z))e^z$ fixed; the thresholds B_{pd} and B_{fd} are increasing in γ_0, in κ and in q, decreasing in $\delta + \lambda$, and, when $d_{\text{first}} < 1$, B_{pd} is increasing in the fixed cost $1 - F(z)$. The thresholds are a calculation; their value is that they make the regions of default explicit in the primitives and the price, which the discrete-time model obtains only numerically.

The thresholds are stated at a given price; in equilibrium the no-default region is $\{B \leq B_{\text{pd}}(z, q(B, z))\}$, a fixed-point condition, which is explicit at the risk-free price. Two implications follow directly, and both are testable in principle. First, a sovereign never defaults a little: the first default is the jump d_{first}, which depends on the cost parameters only, and at the calibrated parameters it is half of the debt service, $d_{\text{first}} \approx 0.52$ at average productivity. The jump is large not because the fixed cost is large but because the variable cost is steep: d_{first} is the ratio of the fixed cost to the variable cost parameter, here 0.13, raised to the power $1/\gamma_1$ with $\gamma_1 = 3.1$. The fixed cost is what makes full repayment possible rather than what produces the jump: with $F(z) = 1$ the variable cost has zero marginal cost at $d = 0$, so $\Gamma(B, z, q) > 0$ and (16) fails at every $B > 0$, the sovereign defaults a little at every positive debt level, and there is no region of full repayment. Any positive fixed cost restores that region, and a jump at its edge is then the only way to leave it. Second, at the risk-free price the no-default region at average productivity ends at $B_{\text{pd}}(0, q^{\text{rf}}) = 0.485$, which is where the steady state of the deterministic model sits (Theorem 1): the debt capacity of the partial default model is an explicit function of the primitives.

Proposition 3 (the bond price)**.** *In the general model, fix measurable policies* (d, S) *and let* (B_t, z_t) *be the induced process. The bounded solution of* (12) *is* (11)*, and it satisfies* $0 \leq q \leq q^{\text{rf}}$*. Moreover* $q(B, z) = q^{\text{rf}}$ *if and only if the sovereign never defaults along the path from* (B, z)*, and* $q(B, z) = 0$ *if and only if it is in full default forever.*

Proposition 4 (continuity of the price along crossing flows)**.** *In the general model, fix a state (B_0, z_0) and suppose that on a neighborhood of it the equilibrium drift is negative and bounded away from zero, so that from every nearby state debt moves down through B_0 in a time that vanishes with the distance to B_0. Then $q(\cdot, z_0)$ is continuous at B_0: with no further condition in the deterministic and chain models, and in the diffusion model whenever $q(B, \cdot)$ is continuous at z_0 uniformly in B on a neighborhood of B_0. The same holds with positive drift. Hence q can be discontinuous in B only at zeros of the drift that are not reached in finite time from both sides (and, in the diffusion model, where q is not continuous in z).*

Proposition 4 is the mechanism of the price cliffs of Section 3.2: at a stationary point of debt that is approached from below and left from above, the paths on the two sides go to different places, the proposition no longer applies, and the price can jump, as it does in the computed solutions. In an equilibrium whose stationary points are all reached in finite time from both sides the price is continuous everywhere.

Proposition 5 (autarky and its removal)**.** *In the general model with productivity in a bounded set: under a zero price, full default is optimal if and only if B is at least $B_{\mathrm{fd}}(z, 0)$ and at least $B_{\mathrm{pd}}(z, 0)$, and $(q \equiv 0, d \equiv 1)$ is an equilibrium on any invariant set contained in $\{B \geq \max\{B_{\mathrm{fd}}(z, 0), B_{\mathrm{pd}}(z, 0)\}\}$; invariance requires the sovereign to issue at least aB at a zero price. Under the convention that no bonds are traded at a zero price, $q > 0$ at every state in every equilibrium.*

Finally, the value is monotone in debt when the price is, with buybacks and with the standard; Proposition 8 in Section 3.2 uses this for the sign of V_B.

Proposition 6 (the value is nonincreasing in debt)**.** *In the general model, if for every z the price $q(\cdot, z)$ is nonincreasing and strictly positive, and the drift cap is slack along the optimal path and along the mimicking path of the proof, then $V(\cdot, z)$ is nonincreasing, with or without buybacks and with or without the underwriting standard.*

3.2 High Debt and the Underwriting Standard

This subsection concerns the model at high debt, where it is not well posed without the borrowing limit, and what the underwriting standard does there. Two things happen in the model at high debt, and the discrete-time literature encounters them together with a third, numerical one. The first is the knife-edge autarky equilibrium of Section 3.1, which a convention removes. The second is an accumulation of debt into the borrowing limit where default carries no fixed cost: the sovereign in full default issues at any positive price up to the limit, whatever its level, so the limit, normally an innocuous parameter, shapes equilibrium behavior at high debt; its entry point looks like a second steady state in computed solutions. We describe it here and remove it with the underwriting standard. The third, a resolution artifact of first-order schemes that mimics a second basin of attraction, is described in Appendix B.3. The numerical evidence behind this subsection, at the parameters of the first of the two calibrations of Section 4.2, is in Appendix C; at those parameters the fixed-cost threshold sits well inside the productivity range. At the final

parameters of Table 1 it sits at the lower edge of the productivity grid, with only the lowest node below it; without the standard the accumulation is present there but carries a negligible share of the ergodic mass, and with it in place the standard binds in partial default only (Section 4.3 and Appendix C).

The free-default accumulation. The mechanism is an argument. In the free-default region a sovereign in full default pays nothing, bears only the variable cost, and faces a marginal cost of debt $-V_B$ close to zero, while the price is positive because productivity will recover; by (9) it issues at any positive price. Even without the borrowing limit it could not keep doing so forever: debt that stayed above every full-default threshold would never be serviced again and would be worthless, which the convention against trade at a zero price rules out (Proposition 5). The next result makes this precise.

Proposition 7 (debt does not grow without bound)**.** *In the general model with productivity in a bounded set Z and without the underwriting standard, drop the borrowing limit, letting debt range over $[0, \infty)$, impose the convention that no bonds are traded at a zero price, and let $\hat{B} := \sup_{z \in Z} \max\{B_{\mathrm{fd}}(z, q^{\mathrm{rf}}), B_{\mathrm{pd}}(z, q^{\mathrm{rf}})\}$. In every equilibrium, from every state (B, z), debt falls to $\hat{B}$ or below at some date with positive probability.*

So without the borrowing limit debt cannot simply grow without bound; whether an equilibrium exists then, and what it looks like at high debt, is open. With the limit, the accumulation runs into it: where default carries no fixed cost, the sovereign in full default borrows up to $\bar{B}$, whatever its level, and the limit is what stops the growth and keeps the price at high debt positive. A parameter that is normally innocuous then shapes equilibrium behavior at high debt. In models that leave the limit implicit, the top of the debt grid plays this role, and a choice about the computation becomes part of the model. The entry point of the accumulation is a stationary point approached from below (the sovereign borrows up to a self-imposed limit just below the full-default line) and left from above, so Proposition 4 does not apply there. Numerically, in the diffusion model at the first calibration, the sovereign in the free-default region issues a third to a half of output per year up to the top of the grid, and the computed price jumps at the entry point. On the cliff nodes the drift sign alternates between iterations, the residual of the pricing equation stays large and grows under refinement, and the share of the ergodic mass that escapes through the cliff is decided at the scale of the grid and by where the grid ends: it moves the mean partial default and the spread volatility by tens of percent across grids (Appendix C, Table 3). No refinement we tried removes it.

The underwriting standard. With issuance switched off below $\underline{q}$, the drift beyond the full-default line is at most $-a(d)B < 0$ and the full-default line is crossed downward from both sides; in the deterministic model the price is continuous across it, and in every model any $\bar{B}$ above the full-default thresholds is an exact truncation (Proposition 9): with the standard the level of the borrowing limit is irrelevant, which is what fails without it. Numerically, in the diffusion model at the first calibration, the floor must exceed the highest price at which the unconstrained

sovereign still issues in full default; a lower floor does not remove the accumulation but relocates it to the debt level at which the price falls through the floor (Appendix C, Table 4); there only a floor of 0.7 removes it entirely, and a floor of 0.6 leaves a small accumulation in place. At the calibrated parameters the cutoff is about 0.3, and the floor of 0.6 clears it with a wide margin: no state issues in full default and the chatter at the full-default line disappears; the drift sign can still alternate on the floor locus itself, where the pricing residual of the computed solutions concentrates (Section 4.2).

What the standard does. Three results describe the equilibrium with the standard in place. Its constraint $\ell \leq 0$ depends on d through $\ell = S + a(d)B$, so the separation of Proposition 1 must be re-examined where it binds.

Proposition 8 (the default decision under the standard)**.** *In the general model with the underwriting standard, fix a state with $q = q(B,z) < \underline{q}$, $B > 0$ and $V_B < 0$, and let (d^*, S^*) be the unconstrained maximizer with issuance $\ell^* = S^* + a(d^*)B$.*

(a) If $\ell^* \leq 0$ the standard is slack and Proposition 1 holds verbatim.

(b) If $\ell^* > 0$ and $d^* > 0$ (the unconstrained optimum is in partial default) the constrained maximum has $\ell = 0$, hence $S = -a(d)B$, and the default intensity maximizes

$$J(d) := u\big(\phi(d,z)e^z - (1-d)(\delta+\lambda)B\big) - a(d)\,B\,V_B.$$

At a maximizer $d^{**} \in (0,1]$, the corner $d^{**} = 1$ included,

$$d^{**} = d_{\text{int}}(B,z,\hat{q}), \qquad \hat{q} := \frac{-V_B}{u'(c^{**})} \in (0,q) :$$

the unconstrained rule (10) evaluated at the sovereign's own shadow price of a unit of debt instead of the market price. Since d_{int} is nonincreasing in the price, $d^{**} \geq d_{\text{int}}(B,z,q)$, strictly whenever $d_{\text{int}}(B,z,q) < 1$.

(c) If $\ell^* > 0$ and $d^* = 0$ (the unconstrained optimum repays in full while issuing), repaying in full forces $\ell = 0$ and gives $J(0)$, while defaulting gives at least $J(d^{**})$, with d^{**} the maximizer of J on $(0,1]$ and $\hat{q}$ as in (b); a default may also come with a buyback, if it frees enough resources. The constrained optimum therefore defaults whenever $B > B_{\text{pd}}(z,\hat{q})$, since J is the objective at the shadow price and the threshold falls with the price. The standard can therefore induce a default that the market-price rule does not.

Where the standard binds, a unit of arrears is valued by the sovereign at $\hat{q}$, which is below the market price because the sovereign would like to issue and cannot, and the sovereign defaults more than the market-price rule prescribes. The numerical kernel maximizes over (d,S) jointly and therefore implements the constrained rule; at the parameters of Table 1 the standard binds in

partial default on just under one percent of the ergodic mass, where the default intensity exceeds the market-price rule by two percent of that rule on average and by up to three percent.

Proposition 9 (the standard removes the accumulation)**.** *In the general model with the underwriting standard, in every equilibrium in which $q < \underline{q}$ at every state where the sovereign is in full default:*

(a) in full default the drift is $S \leq -aB < 0$: the full-default region is transient, and in the deterministic model it is left in time at most $a^{-1} \ln\left(B / \max\{B_{\mathrm{fd}}(0,0), B_{\mathrm{pd}}(0,0)\}\right)$;

(b) in the deterministic model the drift has no zero in $\{q < \underline{q},\ B > 0\}$; a stationary point on the floor locus $\{q = \underline{q}\}$ is approached from above by forced repayment and from below by borrowing, and the price is continuous across it whenever it is reached in finite time from both sides;

(c) any borrowing limit $\bar{B} > \hat{B}$, with $\hat{B}$ as in Proposition 7, is an exact truncation: the drift points inward at $\bar{B}$ in every productivity state and the stationary distribution puts no mass on it;

(d) the thresholds of Proposition 2, as functions of the state and the price, are unaffected; in the deterministic no-default-basin equilibrium, where $q \equiv q^{\mathrm{rf}}$ on $[0, B^*]$, so are the no-default region and the steady state of Theorem 1, whenever $\underline{q} < q^{\mathrm{rf}}$. In the stochastic models a state without default can be priced below the floor, since default is expected after future shocks, and there the standard changes the default decision (Proposition 8(c));

(e) the level of the borrowing limit does not affect the equilibrium on the states below it: for any two limits $\bar{B}_2 > \bar{B}_1 > \hat{B}$, the restriction to $[0, \bar{B}_1]$ of an equilibrium with limit $\bar{B}_2$ is an equilibrium with limit $\bar{B}_1$, and every equilibrium with limit $\bar{B}_1$ extends to one with limit $\bar{B}_2$.

Proposition 10 (the standard bounds the support of debt)**.** *In the general model with the underwriting standard, let $q > 0$ everywhere and $B^\sharp := \sup\{B :\ q(B,z) \geq \underline{q} \text{ for some } z\}$. Then in every equilibrium with any $\bar{B} \geq B^\sharp$: whenever $B_t > B^\sharp$ the drift satisfies $S_t \leq -aB_t$, the set $[0, B^\sharp] \times Z$ is invariant, every stationary distribution is supported in it.*

Proposition 9(e) is the precise sense in which the standard makes the level of the borrowing limit irrelevant, and Proposition 10 bounds the support of debt; under the hypothesis of Proposition 9, $B^\sharp \leq \hat{B}$, so by part (e) neither $B^\sharp$ nor the equilibrium on $[0, B^\sharp]$ depends on a limit above $\hat{B}$. Both results concern limits above the full-default thresholds of every productivity state. The computations use a lower limit, $\bar{B} = 1$, below the full-default thresholds of the best states, so there the irrelevance of the limit is a numerical finding: in the diffusion model at the calibrated parameters, the sovereign chooses an inward drift at high debt in every state, the top of the grid carries no mass, and moving the limit from 1 to 1.5 leaves the moments unchanged (Table 5).

3.3 The Deterministic Model

With z fixed, normalized to 0, the equilibrium is a pair of ordinary differential equations. Writing S^*, c^*, d^* for the policies, (8) and (12) read

$$V'(B) = \frac{\rho V(B) - u(c^*)}{S^*}, \qquad q'(B) = \frac{\tilde{\xi}(d^*)q(B) - (1-d^*)(\lambda+\delta)}{S^*}, \tag{19}$$

with $d^* = d^*(B, q(B))$ from Proposition 2, $c^* = (u')^{-1}(-V'/q)$ and $S^* = (c^* - \mathcal{R}^*)/q$. The system is singular where the drift vanishes, and the stationary points of debt organize the solution. We call a zero of the drift a *steady state*. Theorem 1 pins down one of them within the class of no-default-basin equilibria, the no-default steady state B^*; a zero inside the partial-default region would be a self-fulfilling partial-default steady state, discussed at the end of this subsection; and in the chain model a steady state at the risk-free price can also sit below the default threshold, held in place by precaution (Section 3.4).

Definition 2. An equilibrium of the deterministic model is of *no-default-basin type* with steady state $B^* \in (0, \bar{B})$ if $d \equiv 0$ on $[0, B^*]$, $S > 0$ on $[0, B^*)$, $S(B^*) = 0$ and $S < 0$ on $(B^*, \bar{B}]$. It satisfies the *finite-reach condition* if from $B^* + x$ the state reaches B^* in a time $\tau(x) \to 0$ as $x \downarrow 0$.

Theorem 1 (the deterministic steady state in closed form)**.** *In the deterministic model, let (V, q) be a no-default-basin equilibrium satisfying the finite-reach condition. Then $q \equiv q^{\mathrm{rf}}$ on $[0, B^*]$, $V(B^*) = u(1 - \lambda B^*)/\rho$, and*

$$B^* = B_{\mathrm{pd}}(0, q^{\mathrm{rf}}),$$

the largest debt the sovereign is willing to service in full at the risk-free price, given explicitly by (18) with $q = q^{\mathrm{rf}}$.

The steady state is where the impatient sovereign, borrowing at the risk-free price, runs into its own default threshold. We call it the sovereign's *debt capacity*: the counterpart of the debt limit that the binary model pins down only numerically, here a closed form in the primitives, because the default decision that defines it is a static comparison of the gain from the first default with the fixed cost. Capacity is increasing in the default costs γ_0 and $1 - F$, in the recovery κ, since a higher recovery makes withholding more expensive, and in income; it is decreasing in $\delta + \lambda$, and independent of ρ (as long as $\rho > r$), of the curvature of u, of the drift cap and of the borrowing limit. Table 6 confirms the comparative statics on computed solutions. Steady-state consumption is $1 - \lambda B^*$ and the first partial default beyond B^* has intensity d_{first}. On the basin $[0, B^*)$ the sovereign faces the constant risk-free price and borrows up to the steady state, which it reaches in finite time; the approach is governed by a square-root law.

Lemma 2 (the square-root law)**.** *In the deterministic model, let (V, q) be a no-default-basin equilibrium with V twice continuously differentiable on $(0, B^*)$ and S continuous at B^*, and write $\psi := u'/|u''|$ for the absolute risk tolerance and $\bar{c} = 1 - \lambda B^*$. Then as $x \downarrow 0$*

$$S(B^* - x) = \big(2(\rho - \lambda)\,\psi(\bar{c})\big)^{1/2} x^{1/2}\,(1 + o(1)),$$

and the state reaches B^* from the left in finite time.

Proof. On the basin $q \equiv q^{\mathrm{rf}}$ and $d \equiv 0$, so $c = 1 - \lambda B + S$ and the HJB equation reads $\rho V = u(c) + SV'$ with $V' = -u'(c)$. Differentiating in B and substituting the first-order condition gives $SS' = \lambda S - (\rho - \lambda)\,\psi(c)$. As $B \uparrow B^*$ the drift vanishes and $c \to \bar{c}$, so $(S^2)' \to -2(\rho - \lambda)\psi(\bar{c})$, which integrates to the stated law. The time to reach B^* from $B^* - x$ is $\int_0^x dy/S(B^* - y) \propto x^{1/2}$, which is finite. □

The square-root law is the impatient sovereign's approach to a limit it cannot cross at the risk-free price; Theorem 2 gives the approach from the other side.

Beyond B^* the equilibrium is more delicate: the sovereign is in partial default, the price is below q^{rf}, and the drift is negative, so the state returns to B^* from above. The next result describes the approach.

Theorem 2 (the cube-root law at the steady state)**.** *In the deterministic model, let (V, q) be a no-default-basin equilibrium with the finite-reach condition and $d_{\mathrm{first}} < 1$, and suppose that on a right neighborhood of B^* the drift is interior and V is C^1; let u be C^3 near $\bar{c}$ (for the second-order terms) and real-analytic near $\bar{c}$, as CRRA utility is (for the last statement). Write $x = B - B^*$, $\bar{c} = 1 - \lambda B^*$, and*

$$\Pi_0 := (1-\kappa)(\delta+\lambda)\,d_{\mathrm{first}}, \qquad \beta := \Big(3\,a(d_{\mathrm{first}})\,B^*\,\Pi_0\,\frac{u'(\bar{c})}{|u''(\bar{c})|}\Big)^{1/3}, \qquad \alpha := \frac{3\Pi_0}{2\beta}.$$

Then as $x \downarrow 0$,

$$S(B^*+x) = -\beta\,x^{1/3}\,(1+o(1)), \qquad c(B^*+x) = \bar{c} - \beta\,x^{1/3}\,(1+o(1)),$$
$$q^{\mathrm{rf}} - q(B^*+x) = \alpha\,x^{2/3}\,(1+o(1)),$$

and $V'(B^{*+}) = -u'(\bar{c}) = V'(B^{*-})$. More precisely, with $\psi(c) := u'(c)/|u''(c)|$ and $R_B := -(1 - d_{\mathrm{first}})(\delta+\lambda) + a(d_{\mathrm{first}})$,

$$S(B^*+x) = -\beta\,x^{1/3} - \Sigma_1\,x^{2/3} + O(x), \qquad q^{\mathrm{rf}} - q(B^*+x) = \alpha\,x^{2/3} + Q_1\,x + O(x^{4/3}),$$
$$\Sigma_1 = -\frac{1}{4}\Big[2a(d_{\mathrm{first}})B^*\alpha + \beta^2\frac{\psi'(\bar{c})}{\psi(\bar{c})} + \frac{3\psi(\bar{c})\,(R_B+\rho)}{\beta}\Big], \qquad Q_1 = -\frac{\Pi_0\,\Sigma_1}{\beta^2},$$

where $\psi'(\bar{c})/\psi(\bar{c}) = 1/\bar{c}$ for CRRA utility. Thus V is C^1 but not C^2 at the steady state, the drift leaves zero with a vertical tangent, and the price has a vertical tangent from the right. Moreover the right branch exists on a neighborhood of B^*, is unique among branches that stay bounded in the scaled variables $(q^{\mathrm{rf}} - q)/x^{2/3}$ and $-qS/x^{1/3}$ as $x \downarrow 0$ (hence among those with these asymptotics), and is an analytic function of $(B - B^*)^{1/3}$ (Appendix A).

The economics of the law is slow deleveraging. A sovereign just over its capacity does not snap back: it repays slowly at first and faster as it moves away, so a small excess of debt persists for a long time, while even a small excess carries a visibly lower price. Partial default is what

makes this possible; with a binary decision there would be a jump into default instead. The law has a simple origin. On the right branch the HJB equation reads $\rho V = u(c) + w\,u'(c)$ with $w = -qS \geq 0$ the resources devoted to repayment; letting $B \downarrow B^*$ forces $c \to \bar{c}$ and hence $S \to 0$, and differentiating gives $w' \approx M/w^2$, whence $w^3 \approx 3Mx$. The price gap then integrates $q' \approx -\Pi_0/w$ to the two-thirds power. The constant Π_0 is the flow of lost coupons at the first default; the price has a vertical tangent on the right of the steady state because a small excursion into default costs lenders a flow of coupons over a time that is only $O(x^{2/3})$ but at a rate that does not vanish. Numerically, this is why one-sided differences of V on a production grid look like a kink: on a grid of step h the right difference averages V' over $[B^*, B^* + h]$ and differs from $-u'(\bar{c})$ by $O(h^{1/3})$.

Proposition 11 (existence of the no-default-basin equilibrium)**.** *In the deterministic model with the underwriting standard, let $d_{\text{first}} < 1$, let u be real-analytic near $\bar{c}$, as CRRA utility is, let $\underline{q}$ exceed the full-default prices, and let $B^* = B_{\text{pd}}(0, q^{\text{rf}})$ and $\bar{c} = 1 - \lambda B^*$.*

(a) The left branch. *There is a unique drift $S > 0$ on $[0, B^*)$ with $S(B^*) = 0$ that solves $SS' = \lambda S - (\rho - \lambda)\psi(1 - \lambda B + S)$, $\psi := u'/|u''|$. With $q \equiv q^{\text{rf}}$, $d \equiv 0$, $c = 1 - \lambda B + S$ and V given by $V(B^*) = u(\bar{c})/\rho$ and $V' = -u'(c)$, it satisfies the HJB and pricing equations on $[0, B^*]$, as long as $S < \bar{S}$.*

(b) The right branch. *The solution of (19) that leaves B^* along the cube-root expansion of Theorem 2 exists and is unique near B^*. Suppose its maximal continuation keeps $S^* < 0$, $c^* > c_{\min}$ and $|S^*| < \bar{S}$ up to the full-default threshold, and that beyond it buybacks are not optimal, $-V_B \leq qu'(c)$. Then it extends to all of $[B^*, \bar{B}]$. Where the standard binds before full default, on $\{q < \underline{q},\ \ell^* > 0\}$, the system switches to the constrained regime of Proposition 8(b), $S = -a(d^{**})B$ with d^{**} at the shadow price, and consumption steps down at the switch; beyond the full-default threshold the sovereign neither issues nor buys back, debt decays as $B_t = B_0 e^{-at}$, the price is the power $q(B) \propto B^{-\xi(1)/a}$, and V is given by quadrature.*

(c) The equilibrium. *Under the conditions of (a) and (b), the two branches together form a no-default-basin equilibrium with the finite-reach condition on $[0, \bar{B}]$, the only one whose right branch has the cube-root asymptotics of Theorem 2. In it q is continuous on $[0, \bar{B}]$, and V is C^1 below the point where the standard starts to bind.*

The conditions are on an explicit initial-value problem, not on an equilibrium, so they can be checked at any parameters. The one that matters is that the drift of the right branch has no second zero before full default, that is, no steady state in partial default; the computed solution in the illustration has this shape. A second zero would be a self-fulfilling partial-default steady state. The rest of this subsection says what such a steady state is and when it exists.

Self-fulfilling partial default. In the equilibrium of Theorem 1, lenders expect full repayment, the price is q^{rf}, and the impatient sovereign borrows up to B^* and stops there, because one more

unit of debt would start a default that the price would charge for. Suppose instead that lenders expect the sovereign to settle at a higher debt B° and to withhold a constant share d° of its debt service there forever. They then pay at B° the price of a bond that defaults at rate d° forever,

$$q_r(d) := \frac{(1-d)(\lambda+\delta)}{\xi(d)}, \tag{20}$$

and less than q^{rf} at every lower debt, since every path leads to B°. Two things follow, and together they validate the expectation. First, default starts earlier. At a lower price the arrears created by withholding a unit of debt service are worth less, so withholding costs the sovereign less (Lemma 1); the threshold $B_{\mathrm{pd}}(0,q)$ falls with the price (Proposition 2), and the sovereign defaults at debt levels at which, facing the risk-free price, it would repay in full. Second, the sovereign borrows past its old limit. The default is priced, so on the funds it raises the sovereign pays only the risk-free rate: at the steady state its net payments to lenders, debt service less issuance, are $rq^\circ B^\circ$ per unit of time. Since it discounts at $\rho > r$ it wants to borrow, and it stops at B° only because there each further unit of debt lowers the price at which it must roll over the rest; the impatience wedge balances this dilution exactly (Lemma 3). At B° it withholds the share that lenders expected. Nothing happens suddenly. There is no run and no loss of market access: the sovereign stays in the market, borrows, and drifts into a permanent partial default over years, paying an output cost that exists only because lenders expected the default.

Lemma 3 in Appendix A gives the local conditions at such a point: the price is $q_r(d^\circ)$, its slope is pinned by the impatience wedge, $-q'/q = (\rho - r)/(a(d^\circ)B^\circ)$, and the point attracts only if additional debt dilutes the price strongly enough through additional default. These conditions do not select: they admit a continuum of candidates, one for every intensity $d^\circ \geq d_{\mathrm{first}}$. Whether one of them is an equilibrium is a two-point problem: the sovereign must be willing to borrow into partial default, at prices below q^{rf}, all the way from zero debt. Two cases can be settled analytically. The quantity that governs both is the spread lenders charge a sovereign that withholds, forever, the smallest share it ever withholds, the stationary spread at the first default intensity,

$$s_{\mathrm{first}} := (\lambda+\delta)\Big(\frac{1}{q_r(d_{\mathrm{first}})} - 1\Big) = \frac{(1-\kappa)(\lambda+\delta)\, d_{\mathrm{first}}}{1 - d_{\mathrm{first}}}.$$

Proposition 12 (the standard and self-fulfilling partial default)**.** *In the deterministic model with the underwriting standard, every partial-default steady state has a price at or above the floor, $q_r(d^\circ) \geq \underline{q}$. Since q_r is decreasing and $d^\circ \geq d_{\mathrm{first}}$, a self-fulfilling partial-default steady state can exist only if $q_r(d_{\mathrm{first}}) \geq \underline{q}$, that is, only if the stationary spread at the first default does not exceed the spread at the floor,*

$$s_{\mathrm{first}} \leq (\lambda+\delta)\Big(\frac{1}{\underline{q}} - 1\Big).$$

The condition involves neither the discount rate nor the utility function.

The reason is that standing still requires issuance. At a steady state the sovereign must replace

the debt that matures and is not replaced by arrears, $\ell = a(d^\circ)B^\circ > 0$, and the standard forbids issuance below the floor.

A risk-neutral sovereign. With linear utility the sovereign does not smooth consumption, and the existence question has a nearly complete answer. We let issuance be unrestricted in speed, so that the sovereign can issue a block of debt at once, at the price of the state it moves to. Write $F = F(0)$, and define for $d \in [d_{\text{first}}, 1)$ the stationary spread $s(d) := (1-\kappa)(\lambda+\delta)d/(1-d)$, the price $q_\rho(d) := (1-d)(\lambda+\delta)/(\rho + a(d))$ of a bond that defaults at rate d forever, discounted at the sovereign's rate, and

$$
\begin{aligned}
B(d) &:= \frac{\gamma_0 \gamma_1 F\, d^{\gamma_1 - 1}\, \xi(d)}{(1-\kappa)(\lambda+\delta)^2}, \\
\Theta(d) &:= \frac{1-\phi(d,0)}{q_r(d)B(d)} = s(d)\,\frac{1+(\gamma_1-1)(d_{\text{first}}/d)^{\gamma_1}}{\gamma_1}, \\
\Omega(d) &:= \frac{s(d)\,a(d)}{(\gamma_1-1)\xi(d) - \kappa(\lambda+\delta)d'},
\end{aligned}
$$

with $\Omega(d) = \infty$ when the denominator is not positive.

Proposition 13 (self-fulfilling partial default with a risk-neutral sovereign)**.** *Let $u(c) = c$ and $\rho > r$ in the deterministic model, with issuance unrestricted in speed.*

(a) On any interval of debt through which the sovereign moves at a finite speed it is indifferent, $V' = -q$, and the price solves

$$
q'(B) = -\frac{(\rho + a(d))\,\big(q - q_\rho(d)\big)}{a(d)B}, \qquad d = d^*(B,q),
$$

with drift $S = a(d)B\,\xi(d)\big(q_r(d) - q\big)\,/\,\big((\rho + a(d))(q - q_\rho(d))\big)$; debt rises if and only if $q_\rho(d) < q < q_r(d)$.

(b) The partial-default steady states are the points $(B(d), q_r(d))$, $d \in (d_{\text{first}}, 1)$. The one with intensity d attracts if and only if $\rho - r < \Omega(d)$.

(c) If $\Theta(d) \le \rho - r < \Omega(d)$ and $q_r(d) \ge \underline{q}$, and the solution of (a) through $(B(d), q_r(d))$ continues above it with negative drift up to the full-default threshold or the standard, then there is an equilibrium in which the sovereign issues $B(d)$ at once from zero debt and stays there, withholding the share d forever.

(d) In any equilibrium in which debt rises from zero to a partial-default steady state, $\rho - r \ge \Theta^* := \min_{d \ge d_{\text{first}}} \Theta(d)$.

(e) $\Theta(d_{\text{first}}) = s_{\text{first}}$. If $\gamma_1 \le 2$, Θ is increasing and $\Theta^* = s_{\text{first}}$; and $\Omega(d_{\text{first}}) > s_{\text{first}}$ if and only if $(\gamma_1 - 1)\xi(d_{\text{first}}) < \delta$. When both hold and $q_r(d_{\text{first}}) > \underline{q}$, self-fulfilling partial-default steady states exist for every wedge $\rho - r$ in $(s_{\text{first}}, \Omega(d_{\text{first}}))$, under the continuation condition of (c), and none is reached from zero debt for a wedge below s_{first}.

The condition $\Theta(d) \leq \rho - r$ in (c) is a comparison that a sovereign at zero debt can make directly. Issuing $B(d)$ at once brings $q_r(d)B(d)$. From then on the sovereign pays lenders $rq_r(d)B(d)$ per unit of time, the risk-free rate on what it raised, because the default is priced, and it loses $1 - \phi(d, 0)$ of output to the default. Against staying debt-free forever, front-loading is worth $(\rho - r)q_r(d)B(d)$ per unit of time to an impatient sovereign, and the pessimistic steady state is self-fulfilling when that covers the output cost. Part (d) says that no other route from zero debt is cheaper. Part (e) identifies the threshold with the spread the first default implies: at the first default the output cost equals the payments saved net of the arrears created, and at the stationary price that is the spread times the value of the debt. With a more convex variable cost, deeper defaults cost less on average relative to their spread and the threshold can lie below s_{first}. Every intensity that passes the test in (c) gives an equilibrium, so when one self-fulfilling steady state exists a continuum does, and lenders' expectations alone select among them. In the alternative parameterization of Figure 1, where $\gamma_1 = 1.5$ and $(\gamma_1 - 1)\xi(d_{\text{first}}) = 0.075 < \delta = 0.12$, the threshold is $s_{\text{first}} = 47$ basis points and $\Omega(d_{\text{first}}) = 80$, and at a wedge of 210 basis points the risk-neutral steady states fill the debt interval from 0.87 to 1.07.

Curvature. With CRRA utility the sovereign cannot jump and borrows into default gradually. Near the candidate steady states its consumption is nearly constant along the equilibrium path and the price follows the solutions of Proposition 13(a); away from them curvature slows the borrowing. We compute the equilibria by integrating (19) forward from zero debt, over a grid of starting prices and consumption levels, with a step that resolves the fast adjustment of consumption near the candidates. Integrating down from a candidate is ill posed, because a whole family of solutions enters each candidate from below. Figure 1 shows the outcome at the alternative parameterization with $\nu = 2$ and a wedge of 210 basis points. Next to the no-default steady state at $B = 0.70$ there is a self-fulfilling partial-default steady state at $B = 1.00$, at which the sovereign withholds 28 percent of its debt service at a price of 0.90, a spread of 183 basis points. Facing those prices the sovereign borrows from zero debt all the way into partial default, which starts at $B = 0.62$, lower than under the optimistic price because the threshold rises with the price. This steady state is one of a continuum: at this wedge the steady states reached from zero debt fill the interval from 0.93 to 1.02. The smallest wedge at which any is reached is about 140 basis points for $\nu = 2$, against 47 for a risk-neutral sovereign, and it falls as the curvature falls, to 117 for $\nu = 1$, 89 for $\nu = 0.25$ and 76 for $\nu = 0.1$. A sovereign that values smooth consumption gains less from front-loading, so it needs more impatience to borrow into default.

These numbers suggest the general picture, which we have not proved: with curvature, a self-fulfilling partial-default steady state is reached from zero debt only above a threshold that exceeds the risk-neutral threshold Θ^*, rises with the curvature of utility, and approaches Θ^* as the curvature vanishes. At the parameters of the diffusion model's calibration the question does not arise in the deterministic model, for two independent reasons. The stationary spread at the first default, 1,286 basis points, exceeds the floor spread of 1,278, so the standard excludes

partial-default steady states (Proposition 12), if narrowly. And the impatience wedge, 225 basis points, is far below the risk-neutral threshold, 1,286 basis points with these default costs, so a risk-neutral sovereign would not reach such a steady state from zero debt (Proposition 13(d)), and in every case we computed, curvature raised the threshold. What we have not shown is that the no-default-basin equilibrium is the only equilibrium of any shape at these parameters.

Self-fulfilling partial default is the continuous-time form of the slow-moving debt crises of Lorenzoni and Werning (2019). In their model a pessimistic price raises the cost of rolling over the debt, the sovereign borrows into a crisis that unfolds slowly because debt is a stock that moves at the rate of the deficit, and the crisis validates the price. Here the same logic runs through arrears: a pessimistic price raises the default intensity the sovereign chooses at that price, the higher intensity justifies the price, and debt moves at the rate of the arrears. The difference is where the sovereign's willingness to participate enters. In their setting the sovereign must roll over; here it can stay out of the default region instead, and a candidate survives only if borrowing into default at the pessimistic price beats staying out. Proposition 13 makes that comparison exact for a risk-neutral sovereign: borrowing into default is worth it when the gain from front-loading the borrowed funds covers the output cost of the default.

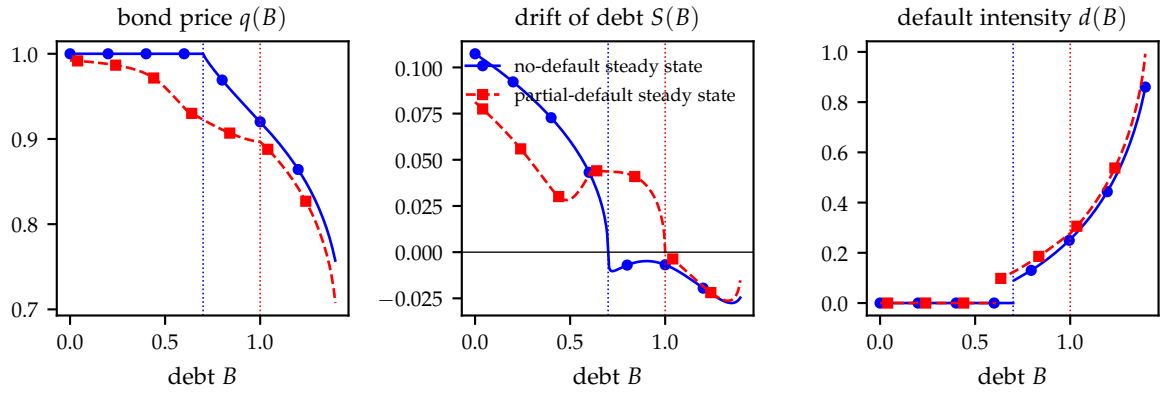


Figure 1: Two equilibria of the deterministic model at an alternative parameterization with a small first default ($\gamma_0 = 0.075$, $\gamma_1 = 1.5$, fixed-cost factor $F(0) = 0.999$, $\rho = 0.06$, $\delta = 0.12$, $\kappa = 0.7$, $r = \lambda = 0.039$, $\nu = 2$, standard at 0.7; first default $d_{\text{first}} = 0.09$, impatience wedge 210 basis points): the no-default steady state at $B = 0.70$ and a self-fulfilling partial-default steady state at $B = 1.00$, where the sovereign withholds 28 percent of its debt service at a price of 0.90. The second is one of a continuum: at this wedge the partial-default steady states reached from zero debt fill the interval from 0.93 to 1.02. Dotted lines: the two steady states. Computed by the branching march of Appendix B.1.

Numerical illustration. Figure 2 shows the deterministic equilibrium in the illustration, with productivity fixed at its mean and the underwriting standard at $\underline{q} = 0.6$, computed by the implicit upwind time stepping of Appendix B on a grid with 500 nodes per unit of debt on $[0, 1.5]$, to residuals of 10^{-7} and 5×10^{-9}. The steady state is at the node 0.480, against $B_{\text{pd}}(0, q^{\text{rf}}) = 0.485$ from (18), and the first default node, 0.484, carries $d = 0.517$ against $d_{\text{first}} = 0.52$ from (17).

Below the steady state the price is q^{rf} and the sovereign borrows; above it the sovereign is in partial default, repays, and the price falls. The underwriting standard binds from $B = 0.888$, where the price falls through 0.6 (dashed vertical line): beyond that point issuance is zero, the drift is $-a(d)B$, and from $B = 1.37$ default is complete and the price follows the power law of Proposition 11. Figure 3 shows the value and consumption: the value is C^1 through the steady state, as Theorem 2 says, while consumption falls steeply at the first default, with the vertical tangent of the cube-root law. Consumption is continuous there: at the threshold the fixed cost equals the debt service saved by the first default, by the definition of the threshold, and what the grid shows as a drop is $\beta h^{1/3}$ for a grid step h. Figure 4 checks Theorem 2: on log-log axes the computed drift and price gap on the right of the steady state, and the drift on its left, are compared with the two-term expansions of the theorem and with the square-root law, with the constants $\beta = 0.172$, $\alpha = 0.54$, $\Sigma_1 = -0.203$ and $Q_1 = 0.425$ computed from the primitives. The leading terms alone describe the solution only within a few grid steps of the steady state: Σ_1/β is -1.2, so the second term of the drift is half the first already at $x = 0.1$, and it is the impatience term $\psi(\bar{c})(R_B + \rho)/\beta$ that makes it large, since the first default is half of the debt service here. With two terms the drift is matched to within seven percent for x between 0.01 and 0.05, and the price gap to within three percent for x between 0.02 and 0.05 and seven percent up to 0.2; further out the next terms of the series take over, and within five grid steps of the steady state the comparison is limited by the grid's placement of the steady state, which the law itself says is off by $O(h)$. The computed drift and price gap leave the steady state with the vertical tangents the theorem predicts.

3.4 The Chain Model

Let productivity take finitely many values z_j and switch from j to k at rate η_{jk}. The HJB and pricing equations are

$$\rho V_j(B) = \max_{c,\,d} \Big\{ u(c) + S\, V_j'(B) + \sum_{k \neq j} \eta_{jk} \big[V_k(B) - V_j(B) \big] \Big\}, \tag{21}$$

$$\xi(d_j)\, q_j(B) = (1 - d_j)(\lambda + \delta) + \tilde{S}_j\, q_j'(B) + \sum_{k \neq j} \eta_{jk} \big[q_k(B) - q_j(B) \big]. \tag{22}$$

Proposition 1 and the thresholds of Proposition 2 hold state by state with $q = q_j(B)$. The chain model is the bridge between the deterministic model, where the sovereign stops exactly at its capacity, and the diffusion model, where it is caught beyond it; its one new element is the precautionary motive, which can hold a steady state at the risk-free price short of capacity.

Proposition 14 (steady states at the risk-free price in the chain model)**.** *In the chain model, suppose $[0, \underline{B}]$ is invariant under the equilibrium dynamics in every state and no state defaults on it. Then $q_j \equiv q^{\mathrm{rf}}$ on $[0, \underline{B}]$ for every j and $\underline{B} \leq \min_j B_{\mathrm{pd}}(z_j, q^{\mathrm{rf}})$. If state j has a steady state at an interior point B° of $[0, \underline{B}]$, at which V_j is twice differentiable, every V_k is differentiable, and the drift cap and the consumption floor*

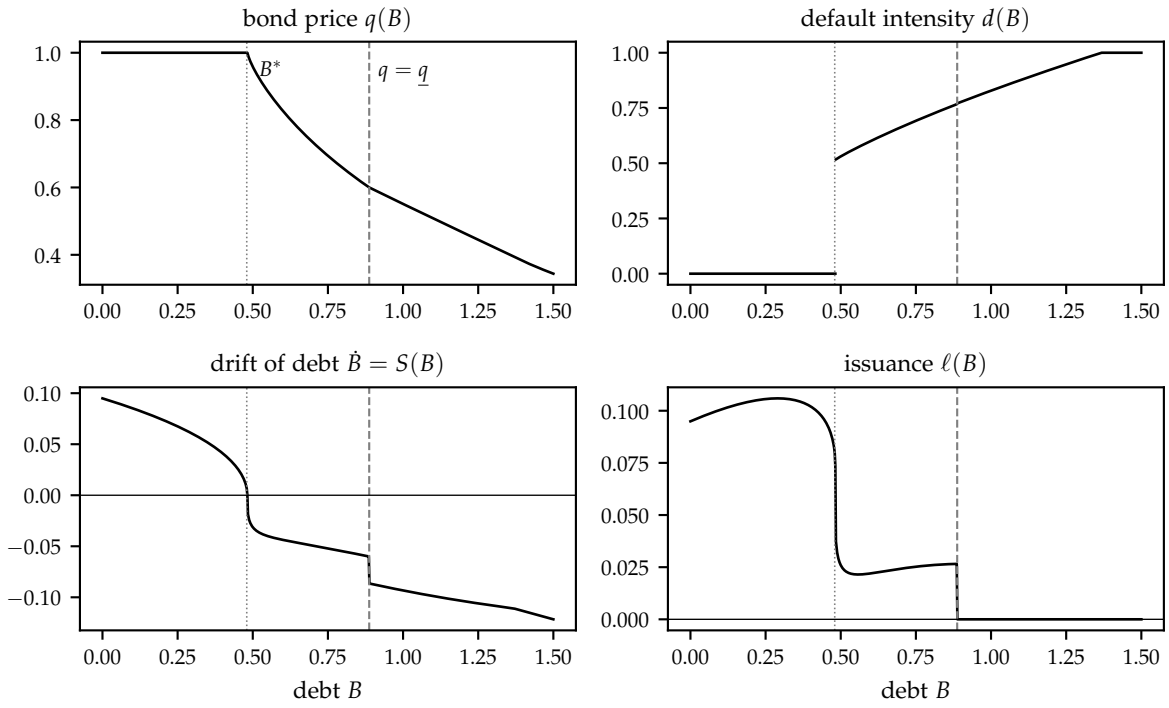


Figure 2: The deterministic model: bond price, default intensity, drift of debt and issuance as functions of debt. Parameters of Table 1 with productivity at its mean; dotted vertical line: the steady state $B^* = B_{\text{pd}}(0, q^{\text{rf}})$; dashed vertical line: the debt at which the price falls through the floor $\underline{q}$.

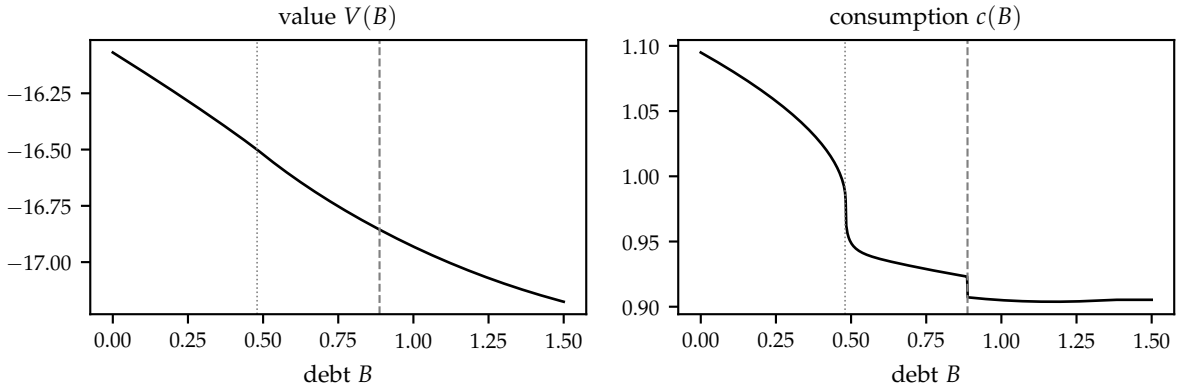


Figure 3: The deterministic model: value and consumption as functions of debt. The value is C^1 at the steady state (Theorem 2); consumption falls steeply but continuously at the first default, with the vertical tangent of the cube-root law, and is nearly flat in the default region, with a small step where the underwriting standard starts to bind (dashed vertical line).

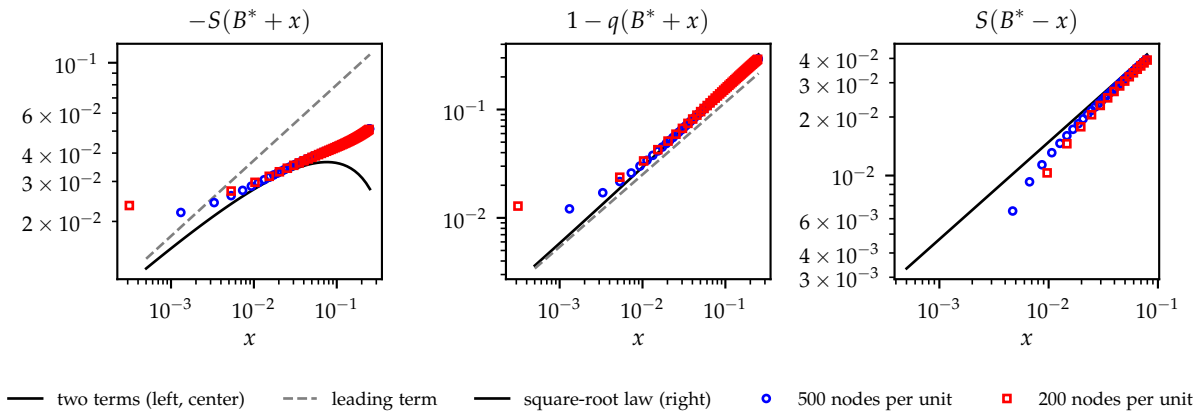


Figure 4: The cube-root law at the steady state. Markers: the computed drift and price gap at a distance x above the steady state (left and center) and the drift at a distance x below it (right), on two grids. Lines: the two-term expansions of Theorem 2 (solid) and their leading terms (dashed), with constants computed from the primitives; right panel, the square-root law.

are slack, then

$$(\rho - \lambda)\, u'(c_j) = \sum_{k \neq j} \eta_{jk} \left(u'(c_k) - u'(c_j) \right) \qquad \text{at } B^\circ. \tag{23}$$

In particular, a state whose consumption at B° is the lowest across states has no steady state in the interior of $[0, \underline{B}]$.

Condition (23) is the Euler equation at a stationary point: the impatience gain from borrowing a little balances the precautionary loss from arriving in a worse state with more debt. In a state whose consumption is the lowest only the impatience term is present, so it cannot rest inside the default-free set and borrows to its edge. The proposition describes chains whose ergodic set is free of default, such as most of the two-state chains discussed next; in the three-state illustration below, where a rare jump to the lowest state leads to default, no such set contains the steady states, and all of them sit at default thresholds. In a better state the two terms compete, and there is a second force that the Euler equation does not show: beyond the threshold of a state that can be reached in one switch, a switch triggers an immediate first default, so the price schedule of the better state has a kink there and the marginal revenue of issuance drops. Numerically, in two-state chains with the default costs of the illustration, this kink pins the high state to the low state's threshold in eight of the nine combinations of shock size and switching rate we examined[5]: both states share a point mass at that debt level, or the high state stops below it, and default does not occur in the ergodic set. The exception is the most persistent and smallest shock, 5 percent with a switching rate of 0.03, where the high state borrows up to its own threshold and default occurs on 4.5 percent of the ergodic mass. The argument suggests that default in the ergodic set requires a switch that lowers the threshold below the current debt, which happens when

5. Nine combinations of shocks of 4, 5, 8 and 11 percent, switching rates of 0.03, 0.05, 0.07 and 0.10, and a discount rate of 0.07 in place of the calibrated 0.062; in six of them the pricing residual stalled between 2×10^{-4} and 2×10^{-2}, including the exception.

a state can jump to a state worse than the one its precautionary motive guards against: a rare direct jump to the bottom, or the continuum of states of the diffusion model, where the threshold moves continuously and the sovereign is caught on the wrong side of it.

Numerical illustration. Figure 5 shows the equilibrium with three states, $z \in \{-0.08, 0, +0.08\}$, in the illustration. Adjacent states switch at the rate $\eta = 0.100$, half the mean-reversion rate $\mu = -\ln \rho_z$ of the diffusion model (Section 4.1), which is the rate that matches the autocorrelation of a symmetric two-state chain; the three-state chain's own productivity decays at the rate η, so its annual autocorrelation is 0.91 against 0.82 for the diffusion, and the two upper states jump directly to the lowest one at the rate $\epsilon = 0.03$, once every three decades. The case is an illustration of the mechanics, not a quantitative exercise: the shock is smaller than the stationary standard deviation of the calibrated process, and the rare jump is the device that lets a chain catch the sovereign on the wrong side of a threshold. The solution is computed on a grid with 201 nodes on $[0, 1]$ with the standard at $\underline{q} = 0.6$, by the time stepping of Appendix B.2, to residuals of 10^{-7} and 3×10^{-9}. The default thresholds are $B = 0.350$, 0.475 and 0.615 in the low, middle and high state, and the default intensity jumps to 0.46 at the first. The middle state borrows up to the node just below its own threshold, $B = 0.470$, and stops there: the rare jump is too unlikely to hold it back, and it is pinned by its own threshold, as in the deterministic model. The high state stops earlier, at the node just below the low state's threshold, $B = 0.345$, where it is pinned by the kink in its price schedule described above; the price there is within one percent of q^{rf}. The low state repays down to its own threshold, the same node, from wherever it finds itself. The ergodic distribution, shown by state in the last panel, has atoms at the two thresholds, mass 0.28 in the middle state at its own threshold and mass 0.16 in the low state and 0.09 in the high state at the low state's threshold, and a continuous part in every state between the two thresholds, traced by the paths that follow a switch. That path is where default happens: a jump from the middle state's atom lands the sovereign in partial default at $B = 0.470$, above the low state's threshold, and it withholds a share of its debt service while it repays. Default occurs on 0.07 of the ergodic mass, all of it in the low state, and mean debt is 0.41. Debt does not rise during these episodes in this illustration: in the low state the sovereign repays from the moment it defaults. In the diffusion model at the calibrated parameters, where the threshold moves continuously with productivity, the sovereign does borrow while in partial default (Section 4.3).

Existence and the price cliffs. Given policies, the prices are the Feynman–Kac values (11) and are continuous except possibly at stationary points of debt that are not reached in finite time from both sides (Proposition 4). In our computations without the underwriting standard the chain model has such a point in the lowest state at the full-default threshold, and the price jumps there; with it, every stationary point of the computed solution is reached from both sides and the prices are continuous. A general existence theorem for the coupled equilibrium is not available: the best-response map from prices to prices is discontinuous (thresholds move with q) and not

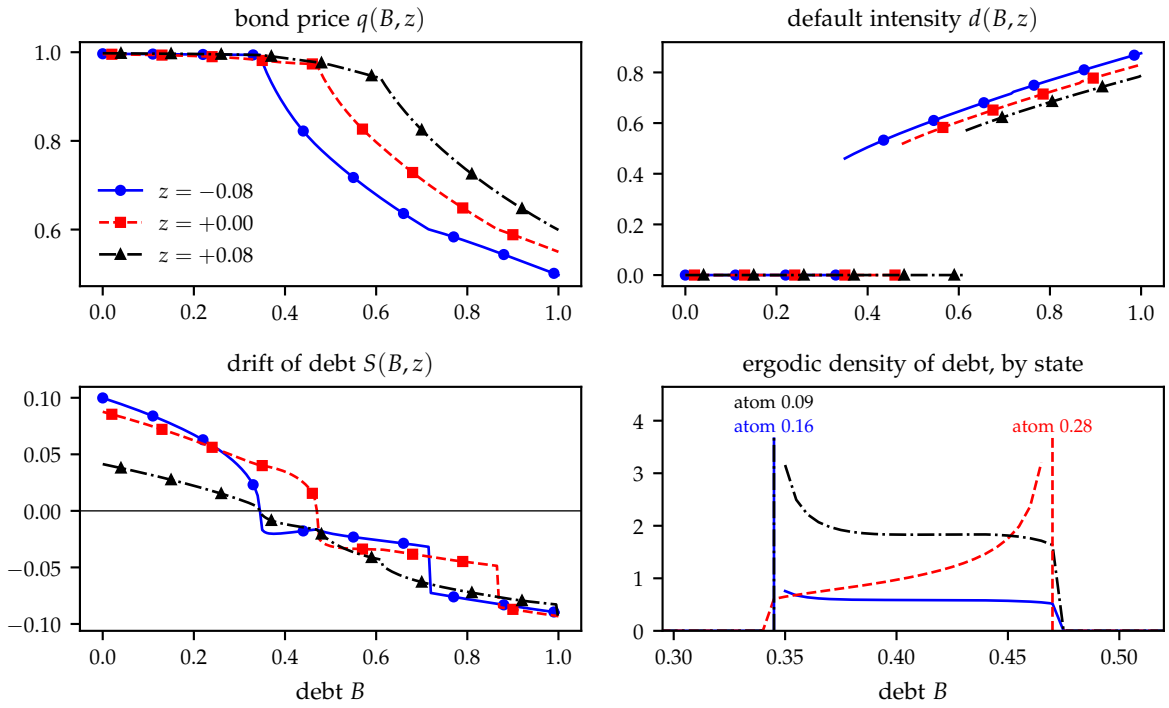


Figure 5: The chain model with three states: bond price, default intensity and drift of debt by productivity state, and the ergodic density of debt by state (atoms are drawn as spikes). Parameters of Table 1; $z \in \{-0.08, 0, 0.08\}$, adjacent switching rate 0.100, rate of the direct jump to the lowest state 0.03.

monotone (a higher price lowers default today but raises borrowing), so neither Schauder's nor Tarski's theorem applies directly. We show in Appendix A (Proposition 15) that the best response does preserve the class of prices nonincreasing in debt, when the induced debt flow is order preserving and the standard is slack.

4 Quantitative Analysis

This section calibrates the diffusion model to the targets of Arellano, Mateos-Planas, and Ríos-Rull (2023) and shows the results of Section 3 at work in the calibrated model: the regions of default and the debt capacity in the policies, the debt targets in the shape of the ergodic set, and the fixed cost in the response to a shock, during which the sovereign in partial default remains a net issuer.

4.1 The Diffusion Model

Log productivity follows an Ornstein–Uhlenbeck process, $dz_t = -\mu z_t\,dt + \sigma\,dW_t$, reflected at $\pm\bar{z}$. Its generator is $\mathcal{L}_z = -\mu z\,\partial_z + \frac{1}{2}\sigma^2\partial_{zz}$, so the HJB and pricing equations are

$$\rho V = \max_{c,\,d}\ \Big\{u(c) + S\,V_B - \mu z\,V_z + \tfrac{1}{2}\sigma^2 V_{zz}\Big\}, \tag{24}$$

$$\xi(d)\,q = (1-d)(\lambda+\delta) + \tilde{S}\,q_B - \mu z\,q_z + \tfrac{1}{2}\sigma^2 q_{zz}, \tag{25}$$

with Neumann conditions $V_z = q_z = 0$ at $\pm\bar{z}$, and the stationary density solves

$$0 = -\partial_B\big[\tilde{S}\,g\big] + \partial_z\big[\mu z\,g\big] + \tfrac{1}{2}\sigma^2\,\partial_{zz}g, \tag{26}$$

with zero-flux conditions $\mu z g + \frac{1}{2}\sigma^2 g_z = 0$ at $z = \pm\bar{z}$ and $\tilde{S}g = 0$ at $B = \bar{B}$, and an atom at $B = 0$ that collects the inflow where $\tilde{S}(0^+, z) < 0$. The process is the continuous-time counterpart of the annual AR(1) of Arellano, Mateos-Planas, and Ríos-Rull (2023): an AR(1) with persistence ρ_z and innovation standard deviation σ_ϵ observed at annual frequency corresponds to $\mu = -\ln\rho_z$ and a stationary standard deviation $\sigma_\epsilon/\sqrt{1-\rho_z^2} = \sigma/\sqrt{2\mu}$. Given a continuous price, (24) is a degenerate elliptic equation with a state constraint in B, whose value function is its unique constrained viscosity solution and is Lipschitz in B (Soner 1986; Capuzzo-Dolcetta and Lions 1990; Fleming and Soner 2006), and the upwind scheme of Appendix B.3 converges to it (Barles and Souganidis 1991). The coupled equilibrium is computed as the fixed point of the two equations.

4.2 Calibration and Computation

Strategy. Utility is CRRA, $u(c) = c^{1-\nu}/(1-\nu)$, and the default cost is (4), the level form of the fixed cost used by Arellano, Mateos-Planas, and Ríos-Rull (2023). We follow their calibration as closely as the continuous-time formulation allows. Risk aversion $\nu = 2$ and the risk-free

rate $r = \ln 1.04$ are fixed. The nine parameters they estimate are our free parameters as well: the default cost parameters $\gamma_0, \gamma_1, \gamma_2, y^*$, the discount rate ρ (their discount factor is $e^{-\rho}$), the recovery κ, the maturity rate δ (their survival rate is $1 - \delta$), and the productivity parameters ρ_z, σ_ϵ mapped as above. The targets are the eleven moments of their Table 4, with their weights (three on the mean of debt due to output, 1.25 on the standard deviation of the spread, one on the rest), and the loss is the weighted sum of squared proportional deviations. The underwriting standard is set at $\underline{q} = 0.6$ (Section 2.4).

Accounting and moments. Debt due is $(\delta + \lambda)B$ per year, debt service is $(1 - d)$ times debt due, defaulted coupons are d times debt due, and the annual partial default is defaulted coupons over debt due within the year, as in their equation (4). The debt level is B (face value, equal to the risk-free value since $q^{\mathrm{rf}} = 1$). Output and consumption are annual integrals, logged and linearly detrended. A year is in partial default if any coupon was defaulted on during it; at the calibrated point the instantaneous probability of being in default under the stationary distribution is 0.27, against an annual frequency of 0.38, the difference being time aggregation. The untargeted moments conditional on default are computed on the simulated annual paths as well. Moments that are functions of the state (debt, debt service and debt due to output, the spread and its correlations) are computed exactly from the stationary distribution of the discretized process, by solving the adjoint system of Appendix B.3. Moments that are path functionals under annual accounting (the partial default frequency, mean and standard deviation, the persistence and volatility of output, consumption) are computed from sixteen simulated paths of 25,000 years each.

Parameters. Table 1 reports the parameters next to those of Arellano, Mateos-Planas, and Ríos-Rull (2023). We calibrate the model in two steps. First without the underwriting standard, to document the behavior at high debt that motivates it at a calibrated point (Section 3.2 and Appendix C); then with the standard in place, which gives Table 1 and everything else in the paper. The calibration under the standard runs twelve Nelder–Mead searches of 300 evaluations each on a coarser productivity grid, continues the four best for 100 evaluations each on the 75-node grid, and reports the best point at which the standard binds on less than one percent of the ergodic mass. The calibrated values differ from theirs in the recovery, which is lower, and in a variable default cost that is higher and more convex, with a fixed cost that rises less steeply with output but starts at a lower output level; the discount rate and the maturity rate are somewhat higher, and productivity is somewhat less persistent, with larger innovations.

Computation. The equilibrium is computed by the implicit upwind finite difference scheme of Achdou et al. (2022), with the value and the price updated jointly at each implicit step because the alternating iteration between them is unstable, on a grid of 201 nodes in debt on $[0, 1]$ and 75 nodes in productivity on ± 2.5 stationary standard deviations. Each parameter evaluation in the

Table 1: Calibration

		AMR (2023)	This paper
Fixed			
ν	risk aversion	2	2
r	risk-free rate ($\ln 1.04$)	0.039	0.039
λ	coupon, $= r$	0.039	0.039
$\underline{q}$	underwriting standard	–	0.60
Calibrated			
γ_0	variable default cost, level	0.0476	0.0759
γ_1	variable default cost, curvature	2.00	3.07
γ_2	fixed default cost, slope	0.120	0.082
y^*	fixed default cost, threshold (level)	0.938	0.751
ρ	discount rate (factor $e^{-\rho}$)	0.047	0.062
κ	recovery of defaulted coupons	0.70	0.373
δ	maturity rate (survival $1-\delta$)	0.12	0.153
ρ_z	annual persistence of productivity	0.875	0.819
σ_ϵ	annual innovation s.d.	0.052	0.066

Notes: The discrete-time parameters of Arellano, Mateos-Planas, and Ríos-Rull (2023) (Table 3) are mapped to continuous time by $\rho = -\ln(\text{discount factor})$, $\delta = 1 - \text{survival rate}$, $\mu = -\ln \rho_z$, $\sigma = \sigma_\epsilon \sqrt{2\mu/(1-\rho_z^2)}$. The free parameters are chosen to minimize the weighted distance to the targets of Table 2, as in their calibration.

calibration solves the model from a warm start in about one minute on a 25-node productivity grid and four minutes on the 75-node grid.[6] At the reported point the HJB residual is 4×10^{-3} and the pricing residual 2×10^{-2}; the latter is carried by nodes on the floor locus $\{q = \underline{q}\}$, where the drift sign alternates between iterations, and the nodes with a pricing residual above 10^{-3} hold about 0.03 percent of the ergodic mass. No taste shocks or other smoothing of the decision problem are needed: the default intensity is a continuous control, the only non-concavity is the fixed cost's jump at $d = 0$, and the monotone three-candidate scheme handles it directly. Discrete-time default models with long-term debt rely on i.i.d. shocks, to income or to the default and borrowing choices, for the price schedule to converge (Chatterjee and Eyigungor 2012; Mihalache 2025); the devices are effective and widely used, and the continuous formulation simply does not need them, which removes both the conceptual distance between the smoothed and the actual decision and the dependence of convergence on the scale of the shocks. Appendix B describes the scheme, the treatment of the non-concave Hamiltonian created by the fixed cost, the initial guess, the convergence criteria, and the exact marching solution of the deterministic model that we use as a check. Section 3.2 describes the difficulties the model poses at high debt and how the underwriting standard resolves them.

6. The calibration searches first on 25 productivity nodes, to economize on computation, and then on 75; every result reported in the paper is computed at the calibrated parameters on the 75-node grid.

4.3 Quantitative Properties

Fit. Table 2 reports the targeted moments in the data, in the model of Arellano, Mateos-Planas, and Ríos-Rull (2023) and in ours, and the untargeted moments of their Table 4. The loss at the current point is 0.077 against 0.228 for their model under the same weights (most of which comes from their standard deviation of debt to output). The continuous-time model reproduces the frequency of partial default, the mean and dispersion of the default intensity in default, the debt service and debt due ratios and the output process; it carries somewhat less debt than the data and generates a spread volatility below the target. Among the untargeted moments, the relative volatility of consumption is close to the data, and defaulted coupons in default are lower and less volatile than in the data, their mean as in their model; the spread is more strongly correlated with output and with debt than in the data, as in their model and more so: in the model the spread is a function of debt and productivity alone, with no shocks on the lenders' side, so it tracks the state more tightly than a spread that also carries global factors.

Table 2: Moments in Data and Model

Moment	Weight	Data	AMR (2023)	This paper
Targeted				
Partial default: frequency	1	0.36	0.37	0.38
Partial default: mean $\mid d > 0$	1	0.38	0.39	0.41
Partial default: s.d. $\mid d > 0$	1	0.22	0.19	0.22
Debt to output: mean	1	0.32	0.32	0.27
Debt to output: s.d.	1	0.18	0.25	0.18
Debt service to output: mean	1	0.036	0.035	0.037
Debt service to output: s.d.	1	0.021	0.022	0.020
Debt due to output: mean	3	0.049	0.050	0.052
Spread: s.d.	1.25	0.041	0.037	0.035
Output: persistence	1	0.89	0.88	0.87
Output: s.d.	1	0.10	0.12	0.11
Loss			0.228	0.077
Untargeted				
Defaulted coupons to output: mean $\mid d > 0$		0.052	0.040	0.040
Defaulted coupons to output: s.d. $\mid d > 0$		0.064	0.036	0.027
Spread: mean		0.053	0.016	0.029
Spread: correlation with output		-0.17	-0.38	-0.78
Spread: correlation with debt		0.24	0.56	0.87
Consumption s.d. relative to output s.d.		1.00	0.91	0.93

Notes: Data and the discrete-time model from Arellano, Mateos-Planas, and Ríos-Rull (2023), Table 4. The loss is the weighted sum of squared proportional deviations from the data.

Values, policies and prices. Figure 6 shows the value, the bond price, consumption, the default intensity and the drift of debt as functions of debt at four productivity levels, and the ergodic marginal distribution of debt; Figure 7 shows the price, the default intensity and the ergodic

density as contour maps over the state space, with the zero-drift locus. The value is decreasing in debt and increasing in productivity. The default policy has the structure of Proposition 2. At the lowest productivity shown, $z = -0.29$, which lies just below the fixed-cost threshold $\ln y^* = -0.286$ (at the calibrated parameters the free-default region contains only the lowest node of the productivity grid, which carries about 0.1 percent of the ergodic mass), the sovereign defaults at any positive debt, as Proposition 2(b) says, with an intensity that rises with debt to nearly full default at the top of the grid; at average and high productivity it does not default until debt reaches the threshold $B_{\mathrm{pd}}(z, q)$, where the default intensity jumps to the first default intensity and rises thereafter. The threshold rises with productivity, so the no-default region is a wedge in the (B, z) plane; full default is not reached on the grid at any productivity level. The price is close to q^{rf} in the no-default region at high productivity and lower at low productivity, where default is expected after future shocks (between 0.89 and 0.99 at zero debt), falls at the default threshold, and declines in the default region, through the floor at high debt; its contours in Figure 7 track the default thresholds. Consumption falls steeply at the default threshold and is nearly flat in debt in the default region, where the sovereign smooths by withholding payments rather than by issuing; the visible steps at low productivity are where the price crosses the floor and issuance stops.

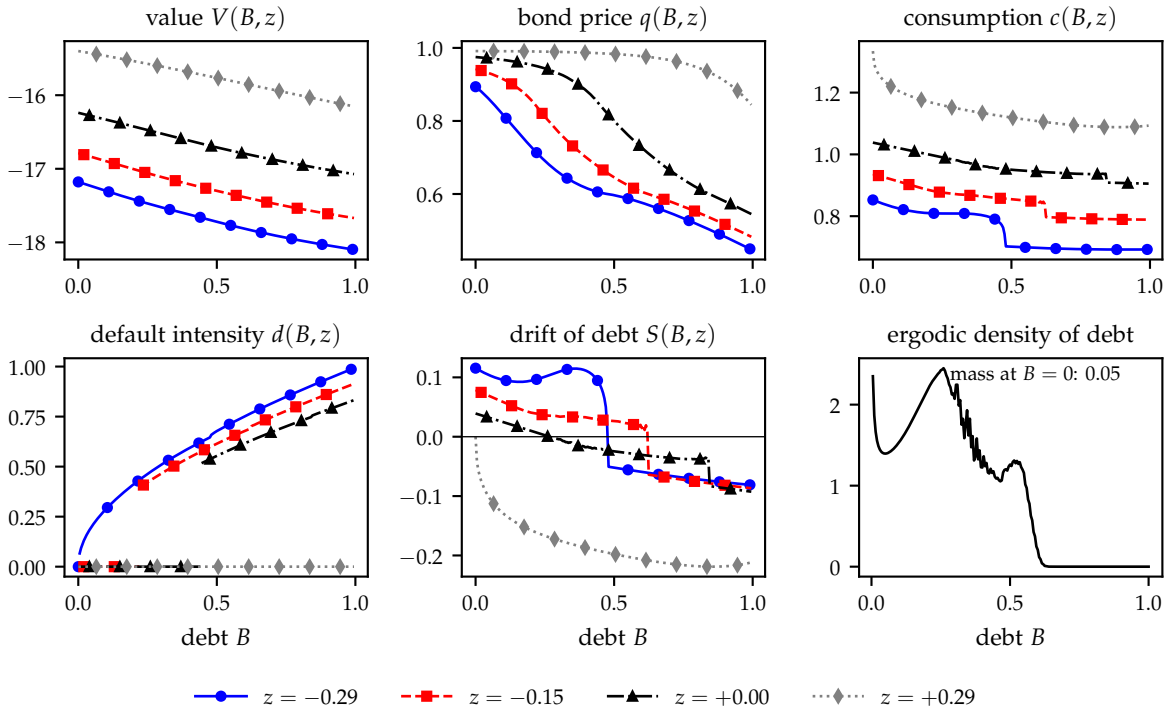


Figure 6: The diffusion model: value, bond price, consumption, default intensity and drift of debt as functions of debt at four productivity levels, and the ergodic marginal density of debt (the atom at $B = 0$ is reported in the panel).

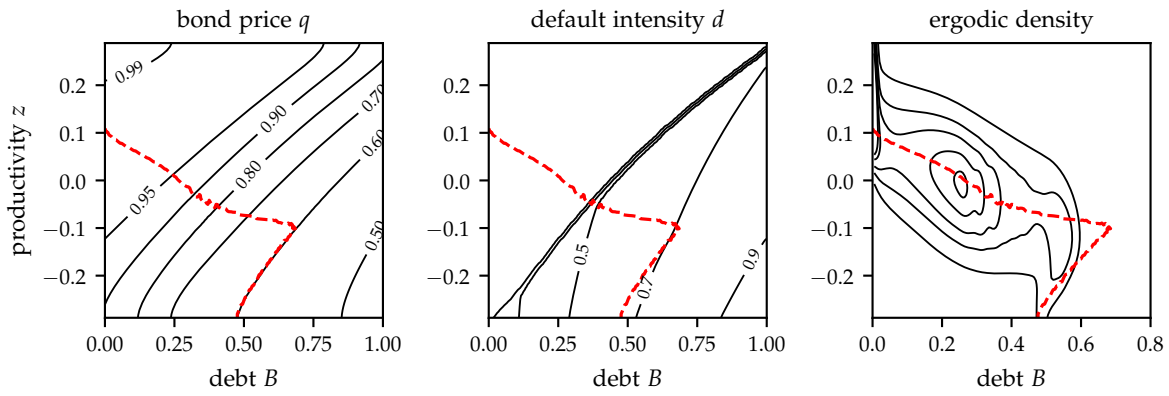


Figure 7: The diffusion model over the state space: contours of the bond price, of the default intensity, and of the ergodic density. Dashed: the zero-drift locus, where debt is stationary.

The drift and the ergodic set. The drift of debt is positive at low debt and negative at high debt in every productivity state below about one standard deviation above the mean, with a single zero-drift locus. Down to about one standard deviation below mean productivity the locus slopes down: the sovereign borrows up to a target debt level that is higher in bad times and lower in good times. In good times the target lies below the no-default threshold: the fixed cost is a brake the sovereign will not cross, and the precautionary motive keeps it short of the threshold. In bad times the threshold has fallen below the current debt, the sovereign is in partial default and pays the fixed cost regardless, and its target is set instead by the dilution of its own price, inside the default region; the episode will end when productivity recovers and the threshold rises back above the debt, not when the debt is repaid, so the sovereign borrows while it lasts. In worse states the target is the debt at which the price reaches the floor and issuance stops, which falls with productivity, so the locus bends back, from $B = 0.67$ to 0.48 at the lowest productivity node. This is the fixed cost at work in the calibrated model: default is lumpy at entry, and sticky after. Above about one standard deviation above mean productivity the drift is negative at every debt level and the sovereign repays all the way to zero debt, which is the atom of the ergodic distribution at $B = 0$. The ergodic set is the region swept by the zero-drift locus as productivity moves: the density in Figure 7 concentrates along the locus, with most of the mass between zero debt and the no-default threshold at average productivity, $B_{\text{pd}}(0, q^{\text{rf}}) = 0.485$, and a tail that extends into the partial-default region at low productivity. It ends well before the top of the grid, at about $B = 0.65$: the full-default region is never reached, and in the worst states the floor locus, where issuance stops, bounds it. Spreads are therefore low most of the time and high in the tail, which is where the model's spread volatility comes from.

Impulse responses. Figure 8 shows the response of the economy to a negative productivity shock. We start from the ergodic distribution, lower z by 0.10 (about one stationary standard deviation, a ten percent fall in output) at date zero, and report the expected path of each variable minus its ergodic mean. The expectation is computed exactly, by evolving the shifted distribution

with the forward equation of the discretized process, so no simulation is involved (Appendix B.3). Output falls on impact and recovers at the rate μ. Consumption falls by less than output and recovers more slowly: the sovereign smooths the shock by withholding debt service, so the default intensity jumps on impact and stays elevated for a decade. Debt *rises* for about six years after the shock, by some six percent of annual output at the peak, before it is worked off; up to the peak, new issues add 8.4 percent of annual output to the debt and arrears 1.6 percent, the rest of the difference being amortization. The spread rises on impact by about 175 basis points and declines with productivity. The last panel decomposes the financing of the shock (new issues in face value and defaulted coupons, both in percent of mean output): on impact the sovereign issues new debt, and new issues exceed defaulted coupons for five and a half years; as debt rises and the price falls, new issues decline, and from then on defaulted coupons are the larger flow while the default intensity is still elevated. These are the dynamics of partial default episodes documented by Arellano, Mateos-Planas, and Ríos-Rull (2023): default is protracted, debt rises during the episode, and the economy exits the episode with more debt than it entered it, as productivity recovers. In the model the sovereign in partial default remains a net issuer: it finances the shock with new issues first and with arrears throughout, stays in the market while it does, and lets the recovery of productivity, not repayment, end the episode.

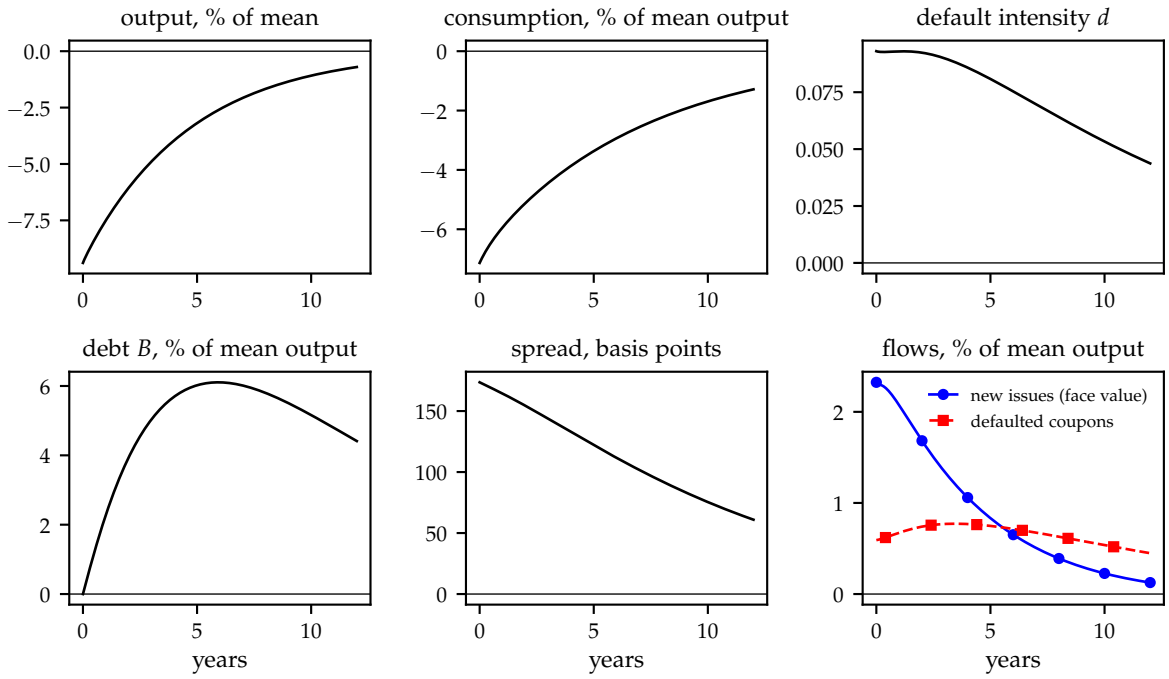


Figure 8: Impulse responses to a negative productivity shock of 0.10 at date zero, from the ergodic distribution: expected path under the forward equation minus the ergodic mean. Flows are in percent of mean annual output; the spread in basis points.

5 Conclusion

Arellano, Mateos-Planas, and Ríos-Rull (2023) replaced the binary default of the standard sovereign debt model with a continuous one and showed that the resulting model matches what default episodes look like. We have studied their model in continuous time, where it is tractable, and asked what the structure of the default costs implies for debt. Three results organize the answer. The sovereign's debt capacity is a closed-form function of the default costs, the recovery on arrears and the maturity of the debt, independent of its preferences, because the default decision is a static comparison of the gain from withholding with the fixed cost, and the first default is a jump of a size the cost parameters determine. Above capacity the sovereign deleverages slowly, along a cube-root law, while the price responds at once. And the model admits self-fulfilling partial default, a permanent default at high debt that a pessimistic price makes cheap enough to borrow into, which for a risk-neutral sovereign requires an impatience wedge at least as large as a threshold we give in closed form, the spread implied by the first default when the default cost is not too convex. Calibrated to the same targets, the diffusion model fits them at least as well, computes fast, exactly and without smoothing devices, and shows the sovereign remaining a net issuer for years into a default episode, because inside the episode the fixed cost is sunk and the debt target is set by dilution instead.

Several avenues remain. On the theoretical side, self-fulfilling partial default is nearly settled for a risk-neutral sovereign, up to an attraction and a continuation condition; with curvature we have computed the threshold but not proved it, and a proof would complete a continuous-time account of slow-moving debt crises with arrears. Existence of the coupled equilibrium in the stochastic models is open for reasons we have made explicit, and so is what the model looks like at high debt without a borrowing limit. On the quantitative side, the implicit price of arrears is set by the recovery parameter and does not respond to the market; a model in which the recovery is the outcome of a renegotiation would make that price depend on the state, and data on arrears and their eventual treatment could discipline it. The natural extensions carry larger state spaces: a measure of small open economies exposed to common global shocks, shocks to the lenders' side, reserves, or a maturity composition. A tensor grid becomes expensive with a third state and infeasible with a fourth, so methods for larger state spaces are the natural next step on the computational side; the closed-form policies and the concentration of the stationary distribution around the thresholds are the features of the model that such methods can exploit.

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Appendix to "Debt Capacity and Slow Deleveraging with Partial Default"

by Sangdong Kim and Gabriel Mihalache

A Proofs and Derivations

A.1 Derivations

Derivation of the HJB equation (8). Over a short interval Δt the principle of optimality gives $V(B_0, z_0) = \max\{u(c_0)\Delta t + e^{-\rho\Delta t}\mathbb{E}_0 V(B_{\Delta t}, z_{\Delta t})\}$. Expanding, $\mathbb{E}_0 V(B_{\Delta t}, z_{\Delta t}) = V + V_B\, S\, \Delta t + \mathcal{L}_z V\, \Delta t + o(\Delta t)$, where $\mathcal{L}_z$ is the generator of z (zero for constant z, $\sum_{k\neq j}\eta_{jk}[V(B, z_k) - V(B, z_j)]$ for a chain, $-\mu z V_z + \frac{1}{2}\sigma^2 V_{zz}$ for the diffusion). Dividing by Δt and letting $\Delta t \to 0$ gives (8).

Derivation of the pricing equation (12). Over $[t, t+\Delta t]$ a unit of debt pays $(1-d_t)(\delta+\lambda)\Delta t$, matures at rate δ, grows by the arrears $\kappa(\delta+\lambda)d_t\Delta t$, and is discounted at r, so $q_t = (1-d_t)(\delta+\lambda)\Delta t + e^{-\xi(d_t)\Delta t}\,\mathbb{E}_t q_{t+\Delta t} + o(\Delta t)$ with $\xi(d) = r + \delta - \kappa(\delta+\lambda)d$. The same expansion gives (12); integrating gives (11).

Derivation of the forward equation (14). For a test function f, $\frac{d}{dt}\mathbb{E}f(B_t, z_t) = \mathbb{E}[\tilde{S} f_B + \mathcal{L}_z f]$, and writing the expectation against the density g and integrating by parts gives $\partial_t g = -\partial_B[\tilde{S} g] + \mathcal{L}_z^* g$, with $\mathcal{L}_z^* g = \partial_z[\mu z g] + \frac{1}{2}\sigma^2\partial_{zz} g$ in the diffusion model. The discrete analogue is exact: if A is the generator matrix of the discretized process, the stationary distribution solves $A^\top g = 0$.

A.2 Proofs for Section 3.1

Proof of Lemma 1. With $\ell = n - \kappa(\delta+\lambda)dB$, (1) gives $c = \phi e^z - (1-d)(\delta+\lambda)B + qn - \kappa q(\delta+\lambda)dB = \phi e^z - (\delta+\lambda)B + qn + (1-\kappa q)(\delta+\lambda)Bd$. At a given drift S, $n = S + \delta B$, so $c = \mathcal{R}(d) + qS$ with $\mathcal{R}$ as stated, and $\mathcal{R}_d$ follows from (4). □

Proof of Proposition 1. Fix any admissible drift S. The objective $u(\mathcal{R}(d) + qS) + SV_B$ depends on d only through $\mathcal{R}(d)$, u is increasing, and the feasibility constraint $c \geq c_{\min}$ is relaxed by a larger $\mathcal{R}(d)$. Hence for every S the best d maximizes $\mathcal{R}$, and so does the d-component of the joint maximizer. □

Proof of Proposition 2. On $(0, 1]$, $\mathcal{R}'(d) = -\gamma_0\gamma_1 F(z)e^z d^{\gamma_1 - 1} + (1-\kappa q)(\delta+\lambda)B$ is strictly decreasing with $\mathcal{R}'(0^+) > 0$, so $\mathcal{R}$ is strictly concave with maximizer d_{int} of (10). The gain from defaulting is $\mathcal{R}(d_{\text{int}}) - \mathcal{R}(0) = \Gamma(B, z, q) - (1 - F(z))e^z$, which gives (16). For an interior d_{int}, $\Gamma = (1 - 1/\gamma_1)(1-\kappa q)(\delta+\lambda)Bd_{\text{int}} \propto B^{\gamma_1/(\gamma_1-1)}$, which is increasing in B, decreasing in q and zero at $B = 0$; (a), (b) and (d) follow. Eliminating B between $\Gamma = (1-F)e^z$ and (10) gives (17), and eliminating d_{int} gives (18). □

Proof of Proposition 3. (11) is the Feynman–Kac representation of (12) with discount rate $\xi(d_t) \geq \xi(1) > 0$. The bound: $(1-d)(\lambda+\delta) \leq q^{\mathrm{rf}}\,\xi(d)$ for all $d \in [0,1]$ is equivalent to $d\,\xi(1) \geq 0$; integrating, $q \leq q^{\mathrm{rf}}\,\mathbb{E}\int_0^\infty \xi(d_t)e^{-\int_0^t \xi}dt = q^{\mathrm{rf}}$, with equality iff $d \equiv 0$. □

Proof of Proposition 4. With the drift bounded away from zero, from (B_0+x, z_0) debt reaches B_0 at a time $\tau(x) \to 0$ on every path along which productivity stays in the neighborhood. By (11) and the strong Markov property, $q(B_0+x,z_0) = \mathbb{E}\big[\int_0^\tau e^{-\int_0^t \xi}(1-d)(\lambda+\delta)\,dt + e^{-\int_0^\tau \xi}\,q(B_\tau,z_\tau)\big]$. The integral is $O(\tau)$ and the discount factor tends to one. In the deterministic model $(B_\tau,z_\tau) = (B_0,z_0)$. For a chain, productivity switches before τ with probability $O(\tau)$, and on the complement $(B_\tau,z_\tau) = (B_0,z_0)$; since q is bounded, the expectation tends to $q(B_0,z_0)$. For the diffusion, productivity leaves the neighborhood before τ with vanishing probability and otherwise $B_\tau = B_0$ with $z_\tau - z_0 = O_p(\tau^{1/2})$, so the expectation tends to $q(B_0,z_0)$ when $q(B_0,\cdot)$ is continuous at z_0. This gives continuity from the right. From the left, start at (B_0,z_0) itself: debt reaches $B_0 - x$ at a time $\tau(x) \to 0$, and the same decomposition gives $q(B_0,z_0) = O(\tau) + \mathbb{E}\big[e^{-\int_0^\tau \xi}\,q(B_0-x,z_\tau)\big]$. In the deterministic and chain models $z_\tau = z_0$ with probability $1-O(\tau)$, so $q(B_0-x,z_0) \to q(B_0,z_0)$. In the diffusion model $q(B_0-x,z_\tau) - q(B_0-x,z_0) \to 0$ in probability by the uniform continuity in z, and the same conclusion follows. With positive drift the two sides exchange roles. □

Proof of Proposition 5. The default rule under $q = 0$ is Proposition 1 with $q = 0$. With $q = 0$ issuance does not enter the budget, so any ℓ is optimal; with $d = 1$ no payment is made and $q = 0$ is the lenders' valuation; invariance is needed for self-consistency, and with $d = 1$ debt decays at rate a unless $\ell \geq aB$. For the removal: suppose $q(B_0,z_0) = 0$. By Proposition 3, $d = 1$ a.e. along the path, so $q = 0$ along the path, so $\ell = 0$ along the path and $B_t = B_0 e^{-at}$. Since $B_{\mathrm{fd}}(z,0) = \gamma_0\gamma_1 F(z)e^z/(\delta+\lambda)$ is bounded below by a positive constant on a bounded productivity set, so is $\max\{B_{\mathrm{fd}}(z,0), B_{\mathrm{pd}}(z,0)\}$, and in finite time B_t falls below $\inf_z \max\{B_{\mathrm{fd}}(z,0), B_{\mathrm{pd}}(z,0)\}$. There full default is not optimal (Proposition 2(d)), so the sovereign pays a positive share of its debt service, a contradiction. □

Proof of Proposition 6. Let $B' < B$ and let (d_t,ℓ_t) be the realized default and issuance path of the optimal policy from (B,z) along the realized z-path, with debt B_t. The sovereign at B' runs the same default path and the issuance $\ell'_t = \ell_t$ if $\ell_t \geq 0$, $\ell'_t = \ell_t\,q(B_t,z_t)/q(B'_t,z_t)$ if $\ell_t < 0$ (the same bonds when issuing, the same expenditure when buying back), with $\ell'_t = 0$ if $B'_t = 0$ and $\ell_t < 0$. First, $B'_t \geq 0$ at all dates: at $B'_t = 0$ the rule gives $\ell'_t \geq 0$, hence $\dot{B}'_t = \ell'_t - a(d_t)B'_t \geq 0$. Next, the gap $b_t = B_t - B'_t$ solves $\dot{b} = (\ell_t - \ell'_t) - a(d_t)b$, with $b_0 > 0$. If b were negative on some interval $(t_2,t_1]$ with $b(t_2) = 0$, then on it $B'_t > B_t \geq 0$, so the override is inactive, and $q(B'_t,z_t) \leq q(B_t,z_t)$ because the price is nonincreasing; hence $\ell'_t = \ell_t$ when $\ell_t \geq 0$ and $\ell'_t \leq \ell_t$ when $\ell_t < 0$, so $\ell_t - \ell'_t \geq 0$. With $-a(d_t)b > 0$ this gives $\dot{b} > 0$ on the interval and $b(t_1) > 0$, a contradiction. Hence $0 \leq B'_t \leq B_t$: the state constraints hold, and the standard is respected because $q(B'_t,z_t) < \underline{q}$ implies $q(B_t,z_t) < \underline{q}$, hence $\ell_t \leq 0$ and $\ell'_t \leq 0$. Consumption satisfies $c'_t - c_t = (1-d_t)(\delta+\lambda)b_t + [q(B'_t,z_t)\ell'_t - q(B_t,z_t)\ell_t] \geq 0$, the bracket being 0 when $\ell_t < 0$, $\ell_t[q(B'_t,z_t) - q(B_t,z_t)] \geq 0$ when $\ell_t \geq 0$, and $-q(B_t,z_t)\ell_t > 0$ when the override is active ($B'_t = 0$

and $\ell_t < 0$). The default costs coincide, so the mimicking path gives the sovereign at B' at least the utility of the optimal path from B. □

A.3 Proofs for Section 3.2

Proof of Proposition 7. Suppose that from (B_0, z_0) debt stays above $\hat{B}$ at every date almost surely. Without the standard the default decision is the static rule of Propositions 1 and 2 at every date. The thresholds B_{pd} and B_{fd} are nondecreasing in the price and $q \le q^{\text{rf}}$ (Proposition 3), so $B_t > \hat{B} \ge \max\{B_{\text{fd}}(z_t, q_t), B_{\text{pd}}(z_t, q_t)\}$ and the sovereign is in full default at every date (Proposition 2(d)). No payment is ever made along the path, and (11) gives $q(B_0, z_0) = 0$, which contradicts Proposition 5, whose argument does not use the upper bound $\bar{B}$. $\hat{B}$ is finite because the thresholds are continuous in z on the bounded set Z. □

Proof of Proposition 8. Parametrize the choice by (d, ℓ). The objective is $\mathcal{G}(d, \ell) = u(\tilde{\mathcal{R}}(d) + q\ell) + (\ell - a(d)B)V_B$ with $\tilde{\mathcal{R}}(d) = \phi(d, z)e^z - (1-d)(\delta+\lambda)B$ the resource function at a zero price, and the constraint is $\ell \le 0$, which no longer involves d. On $(0, 1] \times \mathbb{R}$, $\tilde{\mathcal{R}}$ is concave in d, u is concave and increasing and the last term is linear, so $\mathcal{G}$ is jointly concave. (a) is immediate. For (b), let (d_c, ℓ_c) be a constrained maximizer on the $d > 0$ branch with $\ell_c < 0$; the segment from it to (d^*, ℓ^*) crosses $\{\ell = 0\}$ at a point at which, by concavity, $\mathcal{G}$ is at least $\min\{\mathcal{G}(d_c, \ell_c), \mathcal{G}(d^*, \ell^*)\} = \mathcal{G}(d_c, \ell_c)$, so that point is a constrained maximizer with $\ell = 0$ on the $d > 0$ branch. The $d = 0$ branch is treated separately, since $\mathcal{G}$ is not concave across the fixed-cost jump at $d = 0$: its own unconstrained maximizer has issuance $\ell_0^* > \ell^* > 0$, because $d^* > 0$ maximizes $\mathcal{R}$ and $a(d^*) < \delta$, so the same comparison along $\{d = 0\}$ gives $\ell = 0$ there as well, and the constrained optimum is the better of $J(d^{**})$ and $J(0)$. On $\{\ell = 0\}$, $c = \tilde{\mathcal{R}}(d)$ and $S = -a(d)B$, which gives J. At an interior maximizer $J'(d) = u'(c)\tilde{\mathcal{R}}'(d) + \kappa(\delta+\lambda)BV_B = 0$, that is $\tilde{\mathcal{R}}'(d) = \kappa(\delta+\lambda)B\hat{q}$, the first-order condition of $\mathcal{R}(\cdot; B, z, \hat{q})$; at the corner $d^{**} = 1$, $J'(1) \ge 0$, that is $\tilde{\mathcal{R}}'(1) \ge \kappa(\delta+\lambda)B\hat{q}$, which is $d_{\text{int}}(B, z, \hat{q}) = 1$. In both cases $d^{**} = d_{\text{int}}(B, z, \hat{q})$, and the comparison with $d_{\text{int}}(B, z, q)$ follows from $\hat{q} < q$ and the monotonicity of (10), which is strict below the cap. Finally $\partial_\ell \mathcal{G}(d^{**}, 0) = qu'(c^{**}) + V_B > 0$ because the constraint binds strictly, which is $\hat{q} < q$, and $\hat{q} > 0$ because $V_B < 0$ (Proposition 6). For (c), on the $d = 0$ branch the unconstrained issuance is $\ell^* > 0$, so the comparison of (b) along $\{d = 0\}$ gives $\ell = 0$ and the value $J(0)$. On the $d > 0$ branch the unconstrained issuance can have either sign. If it is positive, the argument of (b) gives $\ell = 0$ and the branch value $J(d^{**})$. If it is nonpositive, the constraint is slack on the branch, and its value $\max_{d \in (0,1], \ell \le 0} \mathcal{G}(d, \ell)$ is at least $\mathcal{G}(d^{**}, 0) = J(d^{**})$. Either way the best default is worth at least $J(d^{**})$. By the concavity of u, $J(d^{**}) - J(0) \ge u'(c^{**})\big[\tilde{\mathcal{R}}(d^{**}) - \tilde{\mathcal{R}}(0)\big] + \kappa(\delta+\lambda)d^{**}BV_B = u'(c^{**})\big[\mathcal{R}(d^{**}; B, z, \hat{q}) - \mathcal{R}(0; B, z, \hat{q})\big]$. Since $d^{**} = d_{\text{int}}(B, z, \hat{q})$ maximizes $\mathcal{R}(\cdot; B, z, \hat{q})$ on $(0, 1]$, the bracket is positive whenever $B > B_{\text{pd}}(z, \hat{q})$ (Proposition 2), so the constrained optimum defaults there. The condition is sufficient, not necessary: the bound uses the concavity of u, and the default region under the standard can extend slightly below $B_{\text{pd}}(z, \hat{q})$. □

Proof of Proposition 9. (a) With $d = 1$ and $\ell \le 0$, $S = \ell - aB \le -aB$; integrate. Full default

requires $B \geq \max\{B_{\mathrm{fd}}(0,q), B_{\mathrm{pd}}(0,q)\}$, and both thresholds are nondecreasing in the price, so in the deterministic model debt has left the full-default region by the time it falls below $\max\{B_{\mathrm{fd}}(0,0), B_{\mathrm{pd}}(0,0)\}$. (b) In $\{q < \underline{q}\}$ the drift is $S \leq -a(d)B \leq -aB < 0$ for every d, so there is no zero there. At a stationary point on the floor locus the paths from the two sides both reach the point; if they do so in finite time, the Feynman–Kac value (11) from either side converges to the value at the point, as in the proof of Proposition 4. (c) At $\bar{B}$ the sovereign is in full default, hence $S < 0$ there in every state, and a state with inward drift carries no stationary mass. (d) The thresholds of Proposition 2 are functions of (B, z, q) and do not involve the restriction. In the no-default-basin equilibrium $q \equiv q^{\mathrm{rf}}$ on $[0, B^*]$ by Theorem 1, and the restriction binds only where $q < \underline{q} < q^{\mathrm{rf}}$, so it is slack on the basin and on a right neighborhood of B^*, where q is continuous. (e) Above $\hat{B}$ the sovereign is in full default in every state at any price $q \leq q^{\mathrm{rf}}$: where the standard is slack by Proposition 2(d), since the thresholds are nondecreasing in the price; where it binds by Proposition 8, since d_{int} is decreasing in the price, the shadow price is below q, and defaulting beats repaying at the shadow price above $B_{\mathrm{pd}}(z, q^{\mathrm{rf}})$. In full default the price is below the floor by hypothesis, so $\ell \leq 0$ and every feasible drift satisfies $S \leq -aB < 0$. Hence from any state with $B \leq \bar{B}_1$ no feasible path, under either limit, rises above $\max\{B, \hat{B}\} \leq \bar{B}_1$. The constraint $B \leq \bar{B}_1$ therefore never restricts the sovereign from these states; given the price on $[0, \bar{B}_1]$, the feasible paths from them, and with them the value and the optimal policies, are the same under both limits; and the price on $[0, \bar{B}_1]$, the Feynman–Kac value (11) along paths that stay in $[0, \bar{B}_1]$, is the same as well. So the restriction of an equilibrium with limit $\bar{B}_2$ is an equilibrium with limit $\bar{B}_1$. Conversely, given an equilibrium with limit $\bar{B}_1$, define the value on $(\bar{B}_1, \bar{B}_2]$ as that of the problem in which the sovereign, in full default, chooses $\ell \leq 0$ until debt reaches $\bar{B}_1$, which it does in time at most $a^{-1}\ln(\bar{B}_2/\bar{B}_1)$, and then continues with the value of the given equilibrium; and define the price there by (11) along the optimal paths, $q(B,z) = \mathbb{E}\left[e^{-\xi(1)\tau} q(\bar{B}_1, z_\tau)\right]$, where τ is the time at which debt reaches $\bar{B}_1$ under the optimal policy of that problem, at most $a^{-1}\ln(B/\bar{B}_1)$ and equal to it when buybacks are not optimal, since no payment is made before τ. This price is below the floor because $q(\bar{B}_1, \cdot) < \underline{q}$, so the standard forbids issuance there, as the construction assumes, and the extended pair is an equilibrium with limit $\bar{B}_2$. □

Proof of Proposition 10. Debt rises only when $\ell > 0$, which the standard permits only at $q \geq \underline{q}$, hence only at $B \leq B^\sharp$; above $B^\sharp$, $\ell \leq 0$ and $S = \ell - a(d)B \leq -aB$. The rest follows from the continuity of B_t. □

A.4 Proofs for Section 3.3

Proof of Theorem 1. From any $B \leq B^*$ the path rises to B^* and stays, with $d \equiv 0$; (11) gives $q = q^{\mathrm{rf}}$ and consumption at B^* is $1 - (\delta + \lambda)B^* + \delta B^* = 1 - \lambda B^*$ forever. At B^* the price is q^{rf} and $d = 0$ is optimal, so $B^* \leq B_{\mathrm{pd}}(0, q^{\mathrm{rf}})$. Suppose $B^* < B_{\mathrm{pd}}(0, q^{\mathrm{rf}})$. By finite reach and (11), $q(B^* + x) \geq e^{-(r+\delta)\tau(x)} q^{\mathrm{rf}} \to q^{\mathrm{rf}}$, and $B_{\mathrm{pd}}(0, \cdot)$ is continuous and increasing, so for small x, $B^* + x < B_{\mathrm{pd}}(0, q(B^* + x))$ and $d = 0$ at $B^* + x$; along the path down to B^* then $d \equiv 0$ and

$q(B^*+x)=q^{\mathrm{rf}}$ exactly. Thus $q\equiv q^{\mathrm{rf}}$ and $d\equiv 0$ on a neighborhood of B^*, where the sovereign faces the constant risk-free price with no default, and staying at B^* is not optimal because $\rho>\lambda$: the round trip that borrows at rate $m=\iota\tau$ for a time τ and repays at the same rate for the next τ perturbs consumption by $\Delta c_t=m-\lambda mt$ on $[0,\tau]$ and $-m-\lambda m(2\tau-t)$ on $[\tau,2\tau]$, with first-order gain $u'(\bar c)\int_0^{2\tau}e^{-\rho t}\Delta c_t\,dt=(\rho-\lambda)u'(\bar c)\iota\tau^3+O(\iota\tau^4)$ and second-order loss $O(\iota^2\tau^3)$; for small ι the gain dominates, contradicting $S(B^*)=0$. □

Proof of Theorem 2. Step 1: the drift vanishes. On the right branch the first-order condition is $V'=-qu'(c)$, and with $S=(c-\mathcal{R})/q$, $\mathcal{R}(B):=\max_d\mathcal{R}(d;B,0,q(B))$, the HJB reads

$$\rho V=u(c)+w\,u'(c),\qquad w:=\mathcal{R}-c=-qS\geq 0. \tag{27}$$

Let $x\downarrow 0$: $V\to u(\bar c)/\rho$, $q\to q^{\mathrm{rf}}$ (by the finite-reach condition and (11), as in the proof of Theorem 1) and $\mathcal{R}\to\bar c$, because at $B^*=B_{\mathrm{pd}}(0,q^{\mathrm{rf}})$ the sovereign is indifferent between $d=0$ and d_{first}. Along any sequence on which $c\to c^+$, (27) gives $u(\bar c)=u(c^+)-(c^+-\bar c)u'(c^+)$; the function $f(c)=u(c)-(c-\bar c)u'(c)-u(\bar c)$ has $f(\bar c)=0$ and $f'(c)=-u''(c)(c-\bar c)$, so $f>0$ for $c\neq\bar c$; hence $c\to\bar c$, $w\to 0$, $S\to 0$, and $V'(B^{*+})=-u'(\bar c)$, which is also the left derivative.

Step 2: the exponents. Put $p:=q^{\mathrm{rf}}-q$. The pricing equation gives $p'=q\Pi/w$ with $\Pi(B,q):=\xi(d^*)q-(1-d^*)(\lambda+\delta)\to\Pi_0>0$. Differentiating (27), using $V'=-qu'(c)$, $c=\mathcal{R}-w$ and the envelope theorem $\mathcal{R}'=\mathcal{R}_B-\mathcal{R}_qp'$ with $\mathcal{R}_B=-(1-d^*)(\delta+\lambda)+qa(d^*)$ and $\mathcal{R}_q=a(d^*)B$,

$$w'=\mathcal{R}_B-\mathcal{R}_qp'+\frac{u'(c)}{|u''(c)|}\,\frac{\mathcal{R}_qq\Pi/w-(\mathcal{R}_B+\rho q)}{w}=\frac{M(x)}{w^2}+O(1/w),$$
$$M(x)\to M:=\frac{u'(\bar c)}{|u''(\bar c)|}\,a(d_{\mathrm{first}})B^*\Pi_0.$$

Since $w\to 0$, $w^2w'=M+o(1)$ and $w^3=3Mx(1+o(1))$, that is $w=\beta x^{1/3}(1+o(1))$ with $\beta^3=3M$. Then $p'=\Pi_0(1+o(1))/w$ integrates to $p=\alpha x^{2/3}(1+o(1))$, and $S=-w/q$, $c=\mathcal{R}-w$ give the stated laws; $V''=-q'u'(c)-qu''(c)c'$ is dominated by $-|u''(\bar c)|\beta x^{-2/3}/3$.

Step 3: the regular system (existence, uniqueness and analyticity of the branch). In the variable $\chi=x^{1/3}$ the scaled unknowns $Q=(q^{\mathrm{rf}}-q)/\chi^2$ and $\Sigma=-qS/\chi$ solve an analytic system of Briot–Bouquet type. With $p=\chi^2Q$, $w=\chi\Sigma$ and $d/dx=(3\chi^2)^{-1}d/d\chi$,

$$\chi Q_\chi=\frac{3q\Pi}{\Sigma}-2Q,\qquad \chi\Sigma_\chi=\frac{3u'}{|u''|}\frac{\mathcal{R}_qq\Pi}{\Sigma^2}-\Sigma-3\chi\Big[\frac{u'}{|u''|}\frac{\mathcal{R}_B+\rho q}{\Sigma}+\frac{\mathcal{R}_qq\Pi}{\Sigma}\Big]+3\chi^2\mathcal{R}_B,$$

with the coefficients evaluated at $c=\mathcal{R}(B^*+\chi^3,q^{\mathrm{rf}}-\chi^2Q)-\chi\Sigma$ and $(B^*+\chi^3,q^{\mathrm{rf}}-\chi^2Q)$. For CRRA utility and $d_{\mathrm{first}}\in(0,1)$ all coefficients are analytic near $(0,\alpha,\beta)$. At $\chi=0$ the right-hand sides vanish exactly at (α,β) and their Jacobian is $\begin{pmatrix}-2 & -3\Pi_0/\beta^2\\ 0 & -3\end{pmatrix}$ (the $(2,2)$ entry is $-6(u'/|u''|)a(d_{\mathrm{first}})B^*\Pi_0/\beta^3-1=-3$ by the definition of β). Neither eigenvalue is a positive integer, so the Briot–Bouquet theorem (Briot and Bouquet 1856; Hille 1976, Chapter 12) yields a

unique solution analytic at $\chi = 0$ with the prescribed value. Since both eigenvalues are negative, every other solution of the linearized system behaves like χ^{-2} or χ^{-3} and is unbounded as $\chi \downarrow 0$, which is the sense in which the branch is unique.

Step 4: the second-order coefficients. The analytic solution has $Q = \alpha + Q_1\chi + O(\chi^2)$ and $\Sigma = \beta + \Sigma_1\chi + O(\chi^2)$. In the coefficients of the system, Π, $\mathcal{R}_q$, $\mathcal{R}_B$, q and d^* differ from their values Π_0, $a(d_{\text{first}})B^*$, R_B, q^{rf} and d_{first} at (B^*, q^{rf}) by $O(\chi^2)$, while $\psi(c) = u'(c)/|u''(c)|$ is evaluated at $c = \bar{c} - \beta\chi + O(\chi^2)$, so $\psi = \psi(\bar{c}) - \psi'(\bar{c})\beta\chi + O(\chi^2)$. Collecting the terms of order χ in the Σ equation, with $3\psi(\bar{c})a(d_{\text{first}})B^*\Pi_0 = \beta^3$, gives $\Sigma_1 = -\beta^2\psi'(\bar{c})/\psi(\bar{c}) - 3\Sigma_1 - 3\psi(\bar{c})(R_B + \rho)/\beta - 3a(d_{\text{first}})B^*\Pi_0/\beta$, which is the formula in the statement after $3a(d_{\text{first}})B^*\Pi_0/\beta = 2a(d_{\text{first}})B^*\alpha$; the terms of order χ in the Q equation give $Q_1 = -3\Pi_0\Sigma_1/\beta^2 - 2Q_1$. Finally $-S = w/q = \chi\Sigma\,(1 + O(\chi^2))$ and $q^{\text{rf}} - q = \chi^2 Q$. □

Proof of Proposition 11. (a) In the parametrization $dB/d\vartheta = S$, $dS/d\vartheta = \lambda S - (\rho - \lambda)\psi(1 - \lambda B + S)$ the basin equation is a regular system, and at $(B^*, 0)$ its vector field is $(0, -(\rho - \lambda)\psi(\bar{c})) \neq 0$, since $\rho > \lambda$. So exactly one trajectory passes through $(B^*, 0)$, and it crosses $S = 0$ downward: for $\vartheta < 0$ it has $S > 0$ and $B < B^*$. Followed backward in ϑ, B falls while $S > 0$, and S cannot return to zero, because on $S = 0$ the field points downward, so the trajectory could only reach $S = 0$ moving forward in ϑ. The system has no rest point with $S \geq 0$, so B reaches 0 at a finite ϑ, and the trajectory is the graph of a drift $S(B) > 0$ on $[0, B^*)$, the unique one with $S(B^*) = 0$. Define c and V as in the statement. Then $\frac{d}{dB}\big(\rho V - u(c) - SV'\big) = -(\rho - \lambda)u'(c) + Su''(c)(S' - \lambda)$, which vanishes by the basin equation, since $S(S' - \lambda) = -(\rho - \lambda)u'/|u''|$; and $\rho V(B^*) - u(\bar{c}) = 0$. So the HJB equation holds on $[0, B^*]$ with the drift S, which is the maximizer given $V' = -u'(c)$ and $q = q^{\text{rf}}$ by concavity, as long as it is below the cap; $d = 0$ is optimal there by Proposition 2, since $B \leq B_{\text{pd}}(0, q^{\text{rf}})$; $S(0) > 0$ satisfies the state constraint; and every path reaches B^* in finite time (Lemma 2) and stays, without default, so (11) gives $q \equiv q^{\text{rf}}$. (b) Local existence and uniqueness near B^* are Step 3 of the proof of Theorem 2, which concerns the system (19) alone. Away from B^* the system (19) is a regular ODE as long as $S^* < 0$, $c^* > c_{\min}$ and $d^* \in (0, 1)$, since $(B, q) \mapsto d^*, \mathcal{R}^*$ are analytic there, so Picard–Lindelöf continues the branch uniquely from the cube-root start up to the first point where one of these fails or the standard starts to bind. On $\{q < \underline{q}\}$ with $\ell^* > 0$ the policy is the constrained one of Proposition 8(b), $S = -a(d^{**})B < 0$, and the HJB equation $\rho V = u(c^{**}) - a(d^{**})BV'$ together with the pricing equation is again a regular system in (V, q), continued in the same way up to the full-default threshold; at the switch consumption drops by $q\ell^*$ and V' may have a kink. Beyond the full-default threshold, with $\underline{q}$ above the full-default prices the sovereign cannot issue, and by hypothesis it does not buy back, so $S = -aB$, the pricing equation $\xi(1)q = -aBq'$ gives the power law, and $V(B) = \int_0^\tau e^{-\rho t}u\big((1 - \gamma_0)F(0)\big)dt + e^{-\rho\tau}V(B_{\text{fd}})$ with $\tau = a^{-1}\ln(B/B_{\text{fd}})$. Continuity of q on $(B^*, \bar{B}]$ follows from Proposition 4, as the drift is negative throughout. (c) The value is C^1 at B^*: on the left $V'(B^{*-}) = -u'(\bar{c})$, and on the right $V' = -qu'(c) \to -u'(\bar{c})$, by Step 1 of the proof of Theorem 2, which uses only the branch. So the HJB equation holds at B^* with $S = 0$, and the pieced pair solves the HJB and pricing equations on $[0, \bar{B}]$, with the policies of the two branches attaining the maximum. It is of no-default-basin

type by construction. It satisfies the finite-reach condition, since by the cube-root law the time to reach B^* from $B^* + x$ is of order $x^{2/3}$. The price is continuous at B^*, which is reached in finite time from both sides, and on each branch. Uniqueness: by Theorem 1 every no-default-basin equilibrium with finite reach has its steady state at B^*; its left branch is the unique trajectory of (a); and a right branch with the cube-root asymptotics is the unique branch of Step 3, continued uniquely as in (b). □

Lemma 3 (local conditions at a partial-default steady state)**.** *In the deterministic model, suppose the equilibrium has a steady state B° in the interior of the partial-default region at which S and q are differentiable and V is twice differentiable, with $d^\circ = d^*(B^\circ, q^\circ) \in (0,1)$. Then $q^\circ = (1-d^\circ)(\lambda+\delta)/\xi(d^\circ)$, the slope of the price is pinned by the impatience wedge,*

$$-\frac{q'(B^\circ)}{q^\circ} = \frac{\rho - r}{a(d^\circ)B^\circ},$$

and the slope of the drift follows from the pricing equation,

$$q'(B^\circ)\big(S'(B^\circ) - \xi(d^\circ)\big) = (\lambda+\delta)(1-\kappa q^\circ)\,\frac{d}{dB} d_{\text{int}}\big(B, q(B)\big)\Big|_{B^\circ},$$

so that the steady state is attracting, $S'(B^\circ) < 0$, if and only if the dilution term on the right exceeds $\xi(d^\circ)\,|q'(B^\circ)|$.

Proof of Lemma 3. The first identity is (11) with constant d. Differentiating $\rho V = u(c) + SV'$ with $c = \mathcal{R}^* + qS$ gives $\rho V' = u'(c)(\mathcal{R}_B + \mathcal{R}_q q' + q'S + qS') + S'V' + SV''$; at $S = 0$ the two terms in S' cancel by the first-order condition $V' = -qu'(c)$, leaving $\rho V' = u'(c)(\mathcal{R}_B + \mathcal{R}_q q')$. With $V' = -qu'(c)$, $\mathcal{R}_q = a(d)B$ and $\mathcal{R}_B = -(1-d)(\delta+\lambda) + qa(d) = -rq$ by the first identity, this reads $-\rho q = -rq + a(d)Bq'$, the second identity. The third is l'Hôpital's rule applied to $q' = (\xi(d)q - (1-d)(\lambda+\delta))/S$. □

Proof of Proposition 12. At a steady state $S = 0$, so issuance is $\ell = S + a(d^\circ)B^\circ = a(d^\circ)B^\circ \geq aB^\circ > 0$, since $a(d) \geq a(1) = a > 0$. The standard (15) forbids $\ell > 0$ when $q < \underline{q}$, so $q^\circ \geq \underline{q}$. The state stays at B° with constant intensity d°, so (11) gives $q^\circ = q_r(d^\circ)$. Differentiating, $q_r'(d) = (\lambda+\delta)\big(\kappa(\lambda+\delta) - (r+\delta)\big)/\xi(d)^2 < 0$, because $\kappa < 1$ and $\lambda = r$; and $d^\circ \geq d_{\text{first}}$, because the static rule never chooses an intensity in $(0, d_{\text{first}})$ (Proposition 2). Hence $\underline{q} \leq q_r(d^\circ) \leq q_r(d_{\text{first}})$. The expression for the spread uses $\xi(d) - (1-d)(\lambda+\delta) = (1-\kappa)(\lambda+\delta)d$, which holds when $\lambda = r$. □

Proof of Proposition 13. With $u(c) = c$ the maximized Hamiltonian is $\mathcal{R}^*(B,q) + S\,(q + V')$, linear in the drift: where $q + V' > 0$ the sovereign issues a block at once, where $q + V' < 0$ it reduces its debt as fast as it can, and a finite positive speed requires $q + V' = 0$.

(a) On an interval with $q + V' = 0$ the HJB equation reduces to $\rho V = \mathcal{R}^*(B, q(B))$. Differentiating and using the envelope theorem, $\mathcal{R}_q^* = a(d)B$, gives $-\rho q = \mathcal{R}_B^* + a(d)Bq'$ with

$\mathcal{R}^*_B = -(1-d)(\lambda+\delta) + qa(d)$, which is the stated equation because $(\rho+a)q - (1-d)(\lambda+\delta) = (\rho+a)(q-q_\rho)$. The pricing equation $\xi(d)q = (1-d)(\lambda+\delta) + Sq'$ gives $S = \xi(d)\big(q - q_r(d)\big)/q'$, and substituting q' gives the drift. Since $\rho > r$, $q_\rho(d) < q_r(d)$, and the sign follows.

(b) At a steady state the price is $q_r(d^\circ)$ by (11), and d° is interior, so $(1-\kappa q)(\lambda+\delta)B = \gamma_0\gamma_1 F d^{\gamma_1-1}$ (Proposition 2); with $(1-\kappa q_r(d))/q_r(d) = (1-\kappa)/(1-d)$ this is $B = B(d)$. At $q = q_r(d)$ the equation in (a) has slope $q' = -q(\rho-r)/(a(d)B)$, the slope of Lemma 3. Write $P(B,q) := \xi(d^*)q - (1-d^*)(\lambda+\delta)$, so that $S = P/q'$ along the solution and $S' = P_q + P_B/q'$ at the steady state. With $P_d = (1-\kappa q)(\lambda+\delta)$, $\partial_B d_{\text{int}} = d/((\gamma_1-1)B)$ and $\partial_q d_{\text{int}} = -\kappa d/((\gamma_1-1)(1-\kappa q))$, we have $P_B = (1-\kappa q)(\lambda+\delta)d/((\gamma_1-1)B)$ and $P_q = \xi(d) - \kappa(\lambda+\delta)d/(\gamma_1-1)$. If $P_q \le 0$ then $S' < 0$; otherwise $S' < 0$ if and only if $\rho - r < P_B a(d)B/(qP_q)$, which is $\Omega(d)$ after using $(1-\kappa q_r)(\lambda+\delta)d/q_r = s(d)$.

(c) Let $B^\circ = B(d)$ and $q^\circ = q_r(d)$. Set $q = q^\circ$ and $V(B) = V(B^\circ) + q^\circ(B^\circ - B)$ on $[0, B^\circ]$, with $\rho V(B^\circ) = \mathcal{R}^*(B^\circ, q^\circ)$, and above B° the solution of (a) and its continuation. From any $B < B^\circ$ the state moves at once to B° and stays, so the lenders' price there is q°, and the issuance is allowed because $q^\circ \ge \underline{q}$. On $[0, B^\circ]$, $q + V' = 0$, so the sovereign is indifferent among speeds and among blocks that end at or below B°, and the HJB equation requires that standing still not be better: $D(B) := \rho V(B) - \mathcal{R}^*(B, q^\circ) \ge 0$. The function D is concave, because $\mathcal{R}^*(\cdot, q)$ is a maximum of functions affine in B; $D(B^\circ) = 0$, and $D'(B^{\circ-}) = (\rho + a(d))\big(q_\rho(d) - q^\circ\big) < 0$. So $D \ge 0$ on $[0, B^\circ]$ if and only if $D(0) \ge 0$. Since $(1-d)(\lambda+\delta) - q^\circ a(d) = rq^\circ$, $\mathcal{R}^*(B^\circ, q^\circ) = \phi(d,0) - rq^\circ B^\circ$ and $D(0) = (\rho - r)q^\circ B^\circ - \big(1 - \phi(d,0)\big)$, which is nonnegative if and only if $\rho - r \ge \Theta(d)$. A block that ends above B°, at B' on the solution of (a), yields $q(B')(B' - B) + V(B')$, which does not exceed $V(B)$ because $V' = -q$ there and q decreases. Above B°, attraction makes the drift negative near B°, and the continuation is the hypothesis. The closed form of Θ follows from $1 - \phi(d,0) = 1 - F + \gamma_0 F d^{\gamma_1}$, from $1 - F = (\gamma_1 - 1)\gamma_0 F d_{\text{first}}^{\gamma_1}$ by (17), and from $q_r(d)B(d) = \gamma_0\gamma_1 F d^{\gamma_1-1}(1-d)/\big((1-\kappa)(\lambda+\delta)\big)$.

(d) Along a path on which debt rises, the sovereign issues blocks or moves along the solutions of (a), and consecutive blocks are one block. Let B_e be the first debt level on the path at which $d^* > 0$. If the path reaches B_e along a solution of (a), the price q_e there satisfies $q_e > q_\rho(0)$, for the drift to be positive just below, where $d = 0$, and $q_e \le q_r(d_{\text{first}})$, for it to be nonnegative just above, where $d = d_{\text{first}}$; so $q_\rho(0) < q_r(d_{\text{first}})$, which is $\rho - r > s_{\text{first}} = \Theta(d_{\text{first}})$. If the path reaches B_e by a block issued from a point $B_O > 0$ of such a solution, the price is continuous at B_O, since a bond held just below B_O is carried into the block, so the landing price q_J equals the price at B_O and exceeds $q_\rho(0)$, while landing with a nonnegative drift requires $q_J \le q_r(d_J) \le q_r(d_{\text{first}})$; again $\rho - r > \Theta(d_{\text{first}})$. If the block is issued at zero debt and lands at (B_J, q_J) with intensity $d_J \in [d_{\text{first}}, 1)$, then $V(0) = q_J B_J + V(B_J)$ with $\rho V(B_J) = \mathcal{R}^*(B_J, q_J)$, and staying at zero debt forever is worth $1/\rho$, so $\mathcal{R}^*(B_J, q_J) + \rho q_J B_J \ge 1$. As $q_J \le q_r(d_J)$, $\mathcal{R}^*(B_J, q_J) \le \phi(d_J, 0) - rq_J B_J$, hence $(\rho - r)q_J B_J \ge 1 - \phi(d_J, 0)$. By the first-order condition for d_J, $\big(1 - \phi(d_J,0)\big)/(q_J B_J) = \big(1 - \phi(d_J,0)\big)(1 - \kappa q_J)(\lambda+\delta)/\big(\gamma_0\gamma_1 F d_J^{\gamma_1-1} q_J\big)$, which is decreasing in q_J and equals $\Theta(d_J)$ at $q_J = q_r(d_J)$; so $\rho - r \ge \Theta(d_J) \ge \Theta^*$.

(e) The closed form gives $\Theta(d_{\text{first}}) = s(d_{\text{first}}) = s_{\text{first}}$. Write $\Theta(d) = (1-\kappa)(\lambda+\delta)\,N(d)/\big(\gamma_1(1-d)\big)$ with $N(d) = d + (\gamma_1 - 1)d_{\text{first}}^{\gamma_1} d^{1-\gamma_1}$; for $d \geq d_{\text{first}}$, $N'(d) = 1 - (\gamma_1-1)^2(d_{\text{first}}/d)^{\gamma_1} \geq 1 - (\gamma_1 - 1)^2 \geq 0$ when $\gamma_1 \leq 2$, and $1/(1-d)$ is increasing, so Θ is increasing. Since $a(d) + \kappa(\lambda+\delta)d = \delta$, $\Omega(d_{\text{first}}) > s(d_{\text{first}})$ if and only if $\delta > (\gamma_1 - 1)\xi(d_{\text{first}})$, including the case of a nonpositive denominator. For a wedge in $(s_{\text{first}}, \Omega(d_{\text{first}}))$, continuity of Θ, Ω and q_r gives an intensity d slightly above d_{first} with $\Theta(d) < \rho - r < \Omega(d)$ and $q_r(d) \geq \underline{q}$, and (c) applies; below s_{first}, (d) excludes them. □

A.5 Proofs for Section 3.4

Proof of Proposition 14. On $[0, \underline{B}]$ no state defaults and every path stays in the set, so (11) gives $q_j = q^{\text{rf}}$ there; no default at that price requires $B \leq B_{\text{pd}}(z_j, q^{\text{rf}})$ in every state (Proposition 2), so $\underline{B} \leq \min_j B_{\text{pd}}(z_j, q^{\text{rf}})$. Near B° the price is $q^{\text{rf}} = 1$ and no state defaults, so consumption in state k is $c_k = e^{z_k} - \lambda B + S_k$, and the HJB equation of state j reads $\rho V_j = \max_S\{u(e^{z_j} - \lambda B + S) + SV_j'\} + \sum_{k\neq j}\eta_{jk}(V_k - V_j)$. The maximand is strictly concave in S and the maximizer is interior, so with V_j twice differentiable at B° the envelope theorem gives, at B°, $\rho V_j' = -\lambda u'(c_j) + S_j V_j'' + \sum_{k\neq j}\eta_{jk}(V_k' - V_j')$, and $S_j(B^\circ) = 0$ removes the second term. The first-order conditions $V_k' = -u'(c_k)$, valid at B° because each V_k is differentiable there and the cap and the floor are slack, turn this into (23). If c_j is the lowest consumption at B°, the right-hand side of (23) is at most zero while the left-hand side is positive, since $\rho > \lambda$. □

Lemma 4 (the price is nonincreasing in the default path)**.** *In the general model, fix a z-path and two default paths with $d_t' \leq d_t$ for all t. Then the Feynman–Kac values* (11) *satisfy $q(d') \geq q(d)$.*

Proof. For $\theta \in [0,1]$ let $d^\theta = d' + \theta(d - d')$ and $Q(\theta)$ the corresponding value. Differentiating under the integral and collecting the effect of d_t^θ on the payment at t and on the discounting of the stream after t (through $\xi'(d) = -\kappa(\delta+\lambda)$), $Q'(\theta) = \mathbb{E}\int_0^\infty e^{-\int_0^t \xi(d^\theta)}(\lambda+\delta)[\kappa q_t^\theta - 1](d_t - d_t')\,dt \leq 0$, since $\kappa q_t^\theta \leq \kappa q^{\text{rf}} \leq 1$. □

Proposition 15 (the best response preserves monotonicity in debt)**.** *In the general model, let q be nonincreasing in B and strictly positive, let the underwriting standard be slack so that the default rule is that of Proposition 1, and suppose the debt flow induced by the optimal policy under q is order preserving (two paths started at $B' < B$ with the same z-path satisfy $B_t' \leq B_t$). Then the price generated by those policies through* (11) *is again nonincreasing in B.*

Proof. By the monotonicity of d^* in B and in q (Section 3.1) and the monotonicity of q, $d^*(B,z) = d^*(B, z, q(B,z))$ is nondecreasing in B; with order preservation, $d^*(B_t', z_t) \leq d^*(B_t, z_t)$ pathwise, and Lemma 4 gives the claim. □

B Numerical Implementation

Discrete equations. Debt lives on a grid B_i, $i = 0, \ldots, n_B - 1$, with step ΔB, and productivity on z_j, $j = 0, \ldots, n_z - 1$. One-sided differences are $D^+V_{ij} = (V_{i+1,j} - V_{ij})/\Delta B$ and $D^-V_{ij} = (V_{ij} - V_{i-1,j})/\Delta B$. At a node, for a given one-sided derivative V_B and price q, consumption is linear in the drift at fixed d, $c(S,d) = \mathcal{R}(d) + qS$, so the Hamiltonian $u(c(S,d)) + SV_B$ is concave in S with maximizer $S = (c^* - \mathcal{R}(d))/q$, $c^* = (u')^{-1}(-V_B/q)$, clamped to the admissible interval (from $c \geq c_{\min}$, $|S| \leq \bar{S}$, the sign restriction of the candidate, and the underwriting standard $S \leq -a(d)B$ where $q < \underline{q}$). Over d the maximized Hamiltonian is concave on $(0,1]$ and is maximized by golden-section search (28 iterations, a tolerance of about 10^{-6} in d), with the candidates $d = 0$, $d = 1$ and the closed form (10) also evaluated; the consumption floor is $c_{\min} = 0.05\,e^z$ and the drift cap $\bar{S} = 1$, and neither binds at any reported solution. Three candidates are formed at every node, forward (D^+V, $S \geq 0$), backward (D^-V, $S \leq 0$) and zero drift; the one with the largest Hamiltonian is the policy. At B_0 the backward candidate is dropped and at B_{n_B-1} the forward one (state constraints). For a concave Hamiltonian this reduces to the standard upwind rule; with the fixed cost of default it is its monotone extension. The z-operator is discretized with upwinding of the exogenous drift and central second differences, so that every off-diagonal coefficient is nonnegative. With A the generator built from the policy, the discrete equilibrium is the root of

$$R_V \equiv \rho V - u(c) - AV = 0, \qquad R_q \equiv \xi(d)\, q - (1-d)(\lambda + \delta) - Aq = 0, \tag{28}$$

and all convergence statements refer to $\|R_V\|_\infty$ and $\|R_q\|_\infty$ at the current (V, q) with the policy recomputed from them, never to step-to-step changes.

B.1 Deterministic Model

The deterministic solutions at the calibrated parameters (Figures 2–4, Table 6) are computed by the time stepping of Appendix B.2 with a single productivity state; Figure 1 by the branching march below, and the continuum of Section 3.3 by forward integration from zero debt. The discrete system also admits an exact solution, which we use as a check and as a tool for the two-point problem of Section 3.3. Nodes with $S < 0$ depend only on the node below, nodes with $S > 0$ only on the node above, and a zero-drift node on nothing, so once the steady-state node is known the system is triangular and can be solved by marching outward. (i) *Candidates.* At each node, all prices q^* solving $q^* = (1 - d^*)(\lambda + \delta)/\xi(d^*)$ with d^* the maximizer of $u(c(0,d))$ at that price are located by a scan of $(0,1]$ with bisection; $V^* = u(c^*)/\rho$. (ii) *Local consistency.* A candidate is kept if for some root of the node above, solved in backward mode, the forward candidate at the steady-state node does not beat zero drift, and symmetrically below. (iii) *Node solve.* Given the upwind neighbor, for each trial q the HJB node equation is monotone in V and solved by bisection, and the price equation becomes a scalar function of q whose roots are all

located. (iv) *Branching march.* March up and down from the candidate, keeping every root that is consistent with the previous node's choice once the new value is known. (v) *Global check.* Verify at every node that the chosen candidate attains the maximum Hamiltonian given both neighbors, and evaluate (28). A solution is accepted if the residuals are below 10^{-8}, exactly one candidate survives with no violation, and the solution on $[0, \bar{B}]$ is unchanged when $\bar{B}$ is increased. At parameters with a small first default the march selects a unique candidate and reproduces the closed-form steady state and the cube-root constants to one or two percent. At the calibrated parameters, where the first default is half of the debt service, every partial-default candidate has a price of at most $q_r(d_{\text{first}}) = 0.599$, just below the floor, so none is an equilibrium under the standard (Proposition 12), and the time-stepped solution agrees with the closed forms to within about two grid steps, four in one case (Table 6). The two-point problem of Section 3.3, whether a candidate partial-default steady state is reached from zero debt, is not a question the march can answer by marching down from the candidate, since a whole family of branches enters it; it is answered by integrating forward from zero debt (Section 3.3).

B.2 Chain Model

The finite-state models are solved by joint implicit upwind time stepping with the policy explicit from (V^k, q^k),

$$\left[(\rho + 1/\Delta t)I - A^k\right]V^{k+1} = u^k + V^k/\Delta t, \qquad \left[(\tilde{\xi}(d^k) + 1/\Delta t)I - A^k\right]q^{k+1} = (1-d^k)(\lambda+\delta) + q^k/\Delta t,$$

with $\Delta t = 0.01$, both solves sparse direct, initialized at the value of servicing the debt at the risk-free coupon forever, $u(e^z - \lambda B)/\rho$, and at the stationary price implied by the interior default policy at the risk-free price, until $\|R_V\|_\infty < 10^{-7}$ and $\|R_q\|_\infty < 10^{-8}$. The joint update matters: the alternating iteration (solve the HJB given q, then the price given the policies) has a loop gain of about thirty at the calibrated parameters and diverges. The upper end of the grid is placed where the drift is negative in every state, so that the truncation is exact; this is checked on the solution. The pricing equation uses the same generator as the HJB equation, so its transport term is upwinded with the equilibrium drift: forward at B_0, where the drift is positive, and backward at B_{n_B-1}, where the state constraint makes it nonpositive. The stationary distribution solves $A^\top g = 0$ with the converged generator.

B.3 Diffusion Model

The diffusion model uses the same time stepping on the two-dimensional grid, with the z-operator discretized as above and Neumann conditions at $\pm\bar{z}$. In the calibration each parameter evaluation runs 3,000 implicit steps of $\Delta t = 0.015$ from a warm start at the previous point, interpolated onto the new z-grid, the first from the same initial guess as the finite-state models; the reported point is the calibrated parameters re-solved on the 75-node productivity grid for a further 9,000 steps, over which the moments are stable to the third digit. Moments are computed

as described in Section 4.2: exactly from $A^\top g = 0$ for state functions, and by Euler–Maruyama simulation with bilinearly interpolated policies, 100 steps per year, annual aggregation of flows and reflecting boundaries for z, for path functionals. The impulse responses of Section 4.3 are expectations under the forward equation $\dot{g} = A^\top g$, started from the ergodic distribution pushed down in productivity by the shock (split linearly between neighboring nodes) and advanced with a Crank–Nicolson step of 0.01 years; a simulation with 20,000 paths gives the same responses to the third digit, with sampling noise visible only in the default intensity, which jumps at the threshold. The ergodic mass on nodes where the resolution ratio (29) below exceeds one half is reported with every solution.

A resolution artifact. The scheme obtains V_B by differencing, with an error of about $\frac{1}{2}|V_{BB}|\Delta B$, and the drift responds to that error with a gain of $1/(q^2|u''(c)|)$: consumption responds to the ratio V_B/q and the drift is the consumption gap divided by q again. Where the *resolution ratio*

$$\frac{\frac{1}{2}|V_{BB}|\,\Delta B}{q^2\,|u''(c)|\,|S|} \tag{29}$$

exceeds about one half, the scheme cannot determine the sign of the drift, and when it loses the sign it tends to produce a positive drift, which carries the state to the top of the grid. This looks like a second basin of attraction but moves with the grid: in the deterministic model the point beyond which the sign is unresolved moved from $B = 2.2$ to 2.6 to beyond 3 as the step was halved twice; in the calibration, a positive drift at average productivity above $B \approx 1.1$ on uniform grids disappeared on a grid refined where the price is small. The remedy is a grid whose spacing is proportional to q^2; we report the ergodic mass on unresolved nodes with every solution.

C Robustness

Grid and barrier, before the standard. Table 3 reports the point calibrated *without* the underwriting standard (the first of the two calibrations of Section 4.2; its parameters are in the notes), solved on four grids, to document the sensitivity that motivated the standard. The bulk is converged: the zero-drift loci at average and high productivity agree across grids to a grid step, and the mass in the debt bands $[0, 0.3)$, $[0.3, 0.5)$ and $[0.5, 0.7)$ is 0.48–0.52, 0.21–0.24 and 0.25 on the refined grids. The tail is not: the mass that escapes into the free-default accumulation region (the top tenth of the grid in Table 3) is 0.9% and 5% on two refined grids ending at $B = 1$ and essentially zero on a refined grid ending at 1.5, because entry happens through a corridor at low productivity whose width is decided at the full-default cliff, where the drift sign chatters. This tail moves the mean partial default between 0.38 and 0.46, debt to output between 0.31 and 0.36, and the spread volatility between 0.022 and 0.058. On these grids the unconstrained sovereign in the free-default region issues at a rate of 0.3 to 0.5 of output per year, the price residual stalls between 0.3 and 0.6 on the cliff nodes, and the highest price at which it still issues in full default

is 0.62 on the uniform grid, which a floor must exceed to remove the accumulation (Table 4).

Table 3: The calibrated point without the underwriting standard on four grids

Grid	Nodes	$\bar{B}$	Loss	Mass top 10%	Mass full default	PD mean	Debt/ output
Uniform, $\Delta B = 0.005$	201	1.0	0.23	0.012	0.017	0.32	0.30
Non-uniform, $h_0 = 0.005$	977	1.0	0.46	0.009	0.023	0.40	0.33
Non-uniform, $h_0 = 0.0025$	1845	1.0	1.25	0.050	0.065	0.46	0.36
Non-uniform, $h_0 = 0.005$	2032	1.5	0.52	0.000	0.003	0.38	0.31

Notes: Parameters of the calibration without the standard: $\gamma_0 = 0.0354$, $\gamma_1 = 2.16$, $\gamma_2 = 0.126$, $y^* = 0.899$, $\rho = 0.0488$, $\kappa = 0.71$, $\delta = 0.124$, $\rho_z = 0.876$, $\sigma_\epsilon = 0.056$ (loss 0.228 on the uniform grid). Non-uniform grids have spacing $h(B) = \max\{h_0 \min_z q(B,z)^2, 2.5 \cdot 10^{-4}\}$. Mass top 10%: ergodic mass in the top tenth of the debt grid.

The level of the standard. Table 4 reports the same point solved with the standard at several levels. Floors of 0.4 and 0.5 do not remove the accumulation but relocate it to the debt level where the price falls through the floor, a trap inside the default region holding two to three percent of the mass; from 0.6 the top of the grid is empty, and at 0.7 no positive drift remains in full default anywhere. At the calibrated parameters of Table 1 the accumulation is much weaker. Solved without the standard on a debt grid extending to $\bar{B} = 2$, the sovereign issues in full default in the lowest productivity states at prices up to 0.29, but the ergodic mass there is about 2×10^{-6}; the moments agree with those of the solution on $[0, 1]$ to within a few thousandths, and the pricing residual stalls at 0.1 on the cliff nodes. At these parameters the standard lowers the spread volatility from 0.042 to 0.035 and raises the loss from 0.065 to 0.077.

Table 4: The calibrated point at several levels of the underwriting standard

$\underline{q}$	Spread at the floor (pp)	Mass in $\{S > 0, d = 1\}$	Max q in $\{S > 0, d = 1\}$	Mass $B \geq 0.9$	Mass full default
none	–	0.013	0.62	0.012	0.017
0.4	24.5	0.030	0.62	0.025	0.059
0.5	16.3	0.020	0.64	0.005	0.040
0.6	10.9	0.003	0.67	0.000	0.032
0.65	8.8	0.000	0.68	0.000	0.023
0.7	7.0	0	–	0.000	0.012

Notes: Uniform grid, $\bar{B} = 1$, parameters of the calibration without the standard. The spread at the floor is $(\delta + \lambda)(1/\underline{q} - 1)$, in percentage points.

Robustness of the calibrated point under the standard. Table 5 re-solves the calibrated point from a warm start on a finer debt grid, on a non-uniform grid of the kind of Appendix B.3, and with the barrier at $B = 1.5$. With the standard in place the moments move by amounts

comparable to the sampling noise of the simulated moments, the top of the grid carries no mass, and the loss is unchanged to the second digit.

Table 5: The calibrated point under the standard on other grids

Grid	Debt/y mean	Debt/y s.d.	Spread s.d.	PD freq.	PD mean	Loss	Mass top	Unresolved	Residuals
main grid	0.269	0.183	0.035	0.378	0.409	0.077	0.0000	0.015	4e-03 / 2e-02
debt step halved	0.269	0.183	0.035	0.376	0.410	0.078	0.0000	0.007	4e-03 / 2e-02
non-uniform debt grid	0.269	0.183	0.035	0.375	0.410	0.078	0.0000	0.005	3e-03 / 2e-02
barrier at $B = 1.5$	0.269	0.183	0.035	0.378	0.409	0.078	0.0000	0.015	3e-03 / 2e-02

Notes: Each row re-solves the calibrated point from a warm start. Main grid: 201 debt nodes (step 0.005) on $[0, 1]$ and 75 productivity nodes; debt step halved: 401 debt nodes; non-uniform: 572 debt nodes with a minimum step of 0.00025; all rows use 75 productivity nodes. Mass top: ergodic mass in the top tenth of the debt grid; unresolved: ergodic mass on nodes where the resolution ratio (29) exceeds one half; residuals: final HJB and pricing residuals.

Comparative statics. Table 6 re-solves the deterministic model across variations of the default cost, the recovery parameter, the discount rate and the maturity around the calibrated point, by the time stepping of Appendix B.2 on 200 nodes per unit of debt on $[0, 1.2]$, and compares the closed forms of Theorem 1 with the computed solution. In every case the last node with positive drift is two nodes below the first default node, and the computed default intensity at the first default node agrees with d_{first} to within 0.004. The first default node is within one grid step of $B_{\text{pd}}(0, q^{\text{rf}})$ in every case but $\kappa = 0.65$, where it lies 1.7 steps below; correspondingly, the computed steady state lies between 1.7 and 2.6 grid steps below $B_{\text{pd}}(0, q^{\text{rf}})$, and 3.7 steps below at $\kappa = 0.65$. The steady state rises with γ_0, κ and γ_2 and falls with δ, as the closed form says and the monotonicity properties of Section 3.1 predict for the default cost and the recovery parameter, and it does not move with ρ: the discount rate changes only the price schedule beyond the steady state, and with it the point where the standard starts to bind; full default is reached on the grid in two cases only.

Table 6: Comparative statics of the deterministic steady state

	Steady state		First default			Standard	Full default
Variation	$B_{\mathrm{pd}}(0, q^{\mathrm{rf}})$	computed	d_{first}	computed	$B_{\mathrm{fd}}(0, q^{\mathrm{rf}})$	binds from	from
Calibrated point	0.485	0.475	0.517	0.519	1.90	0.890	–
$\gamma_0 \times 0.8$	0.451	0.440	0.556	0.557	1.52	0.785	1.11
$\gamma_0 \times 1.2$	0.514	0.505	0.487	0.489	2.28	0.990	–
$\kappa = 0.45$	0.553	0.540	0.517	0.518	2.16	1.005	–
$\kappa = 0.65$	0.869	0.850	0.517	0.517	3.40	–	–
$\gamma_2 \times 0.5$	0.305	0.295	0.411	0.412	1.92	0.710	–
$\gamma_2 \times 1.5$	0.635	0.625	0.592	0.594	1.88	1.040	–
$\rho = 0.05$	0.485	0.475	0.517	0.519	1.90	0.940	–
$\rho = 0.07$	0.485	0.475	0.517	0.519	1.90	0.860	–
$\delta = 0.12$	0.584	0.575	0.517	0.519	2.28	1.020	–
$\delta = 0.22$	0.358	0.350	0.517	0.520	1.40	0.705	1.03

Notes: One parameter changed at a time from the calibrated point (Table 1), productivity at its mean, $\underline{q} = 0.6$. "Computed" steady state: the last grid node with positive drift; "computed" first default: the default intensity at the first node with $d > 0$. Residuals 10^{-7} and 10^{-8} in all cases but one ($\delta = 0.12$), where the price residual stalled at 4×10^{-3} after 30,000 steps. "–": full default is not reached on the grid.